



CS 540 Introduction to Artificial Intelligence

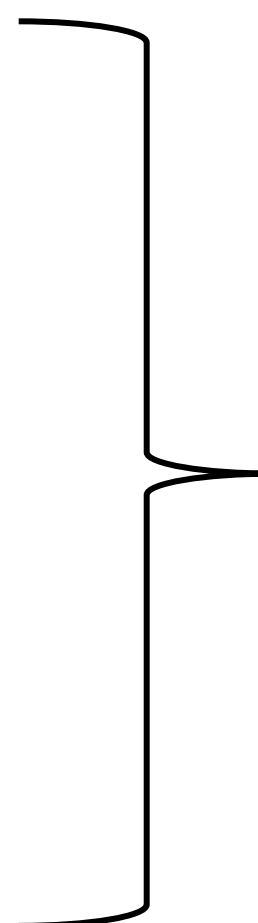
Machine Learning Overview

Spring 2025

Announcements

- **HW3 due on Thursday 13th at 11:59 PM**

Class roadmap:

ML Introduction	 Machine Learning
ML Unsupervised I	
ML Unsupervised II	
ML: Linear Regression	

Today's outline

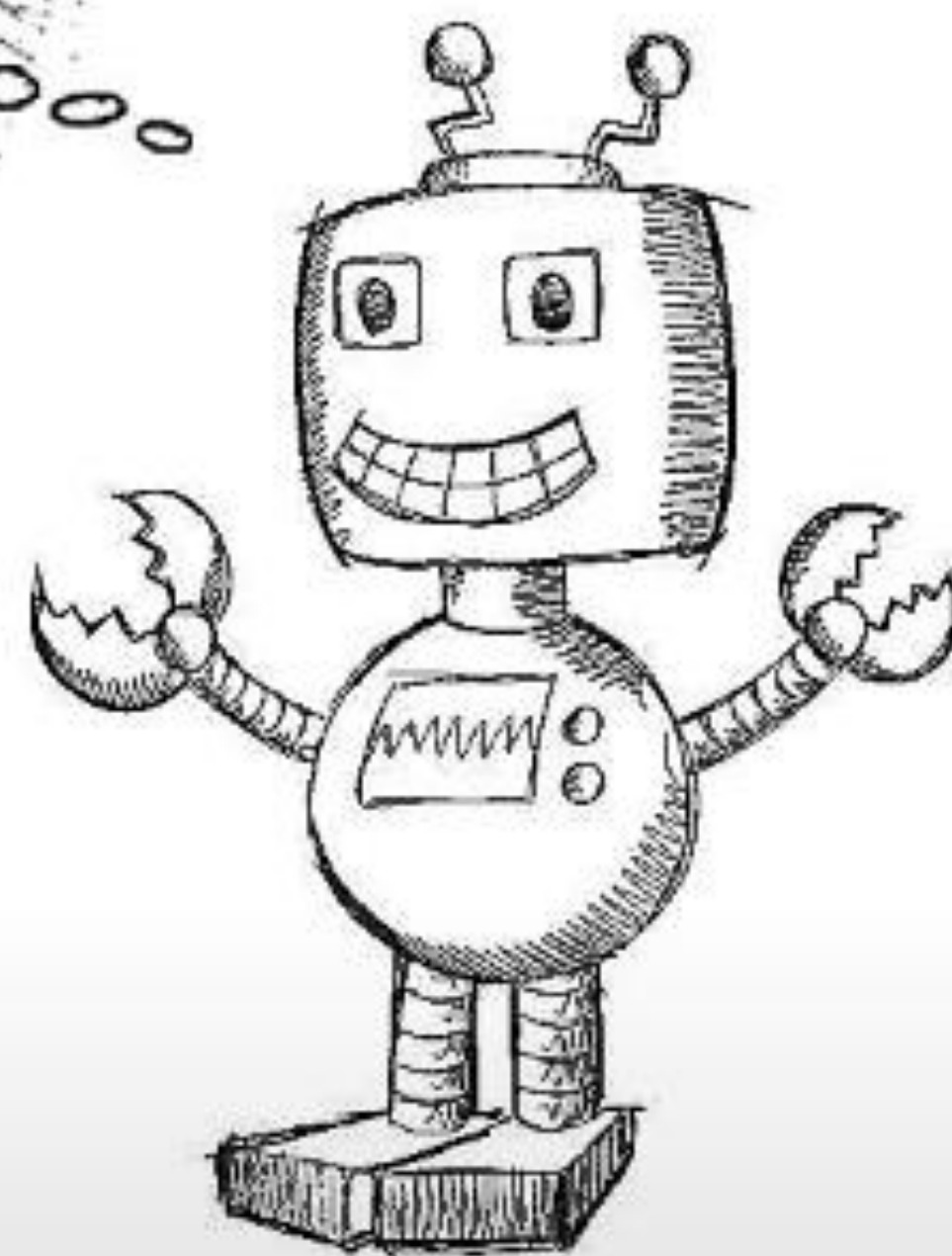
- What is machine learning?
- Supervised Learning
 - Classification
 - Regression
- Unsupervised Learning
 - Clustering
- Reinforcement Learning



Part I: What is machine learning?



**HUMANS LEARN FROM
PAST EXPERIENCES**



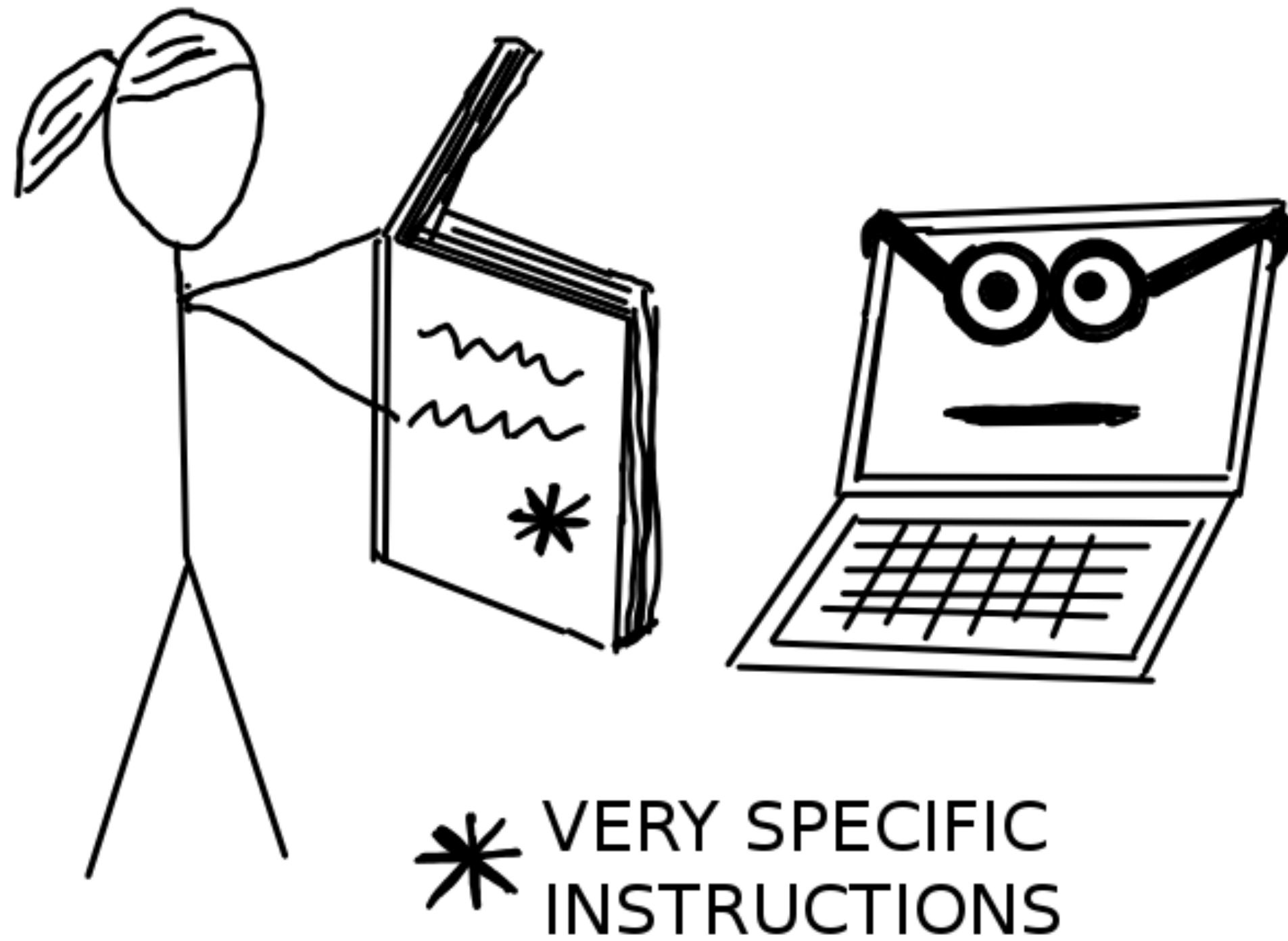
**MACHINES FOLLOW INSTRUCTIONS
GIVEN BY HUMANS**

What is **machine learning**?

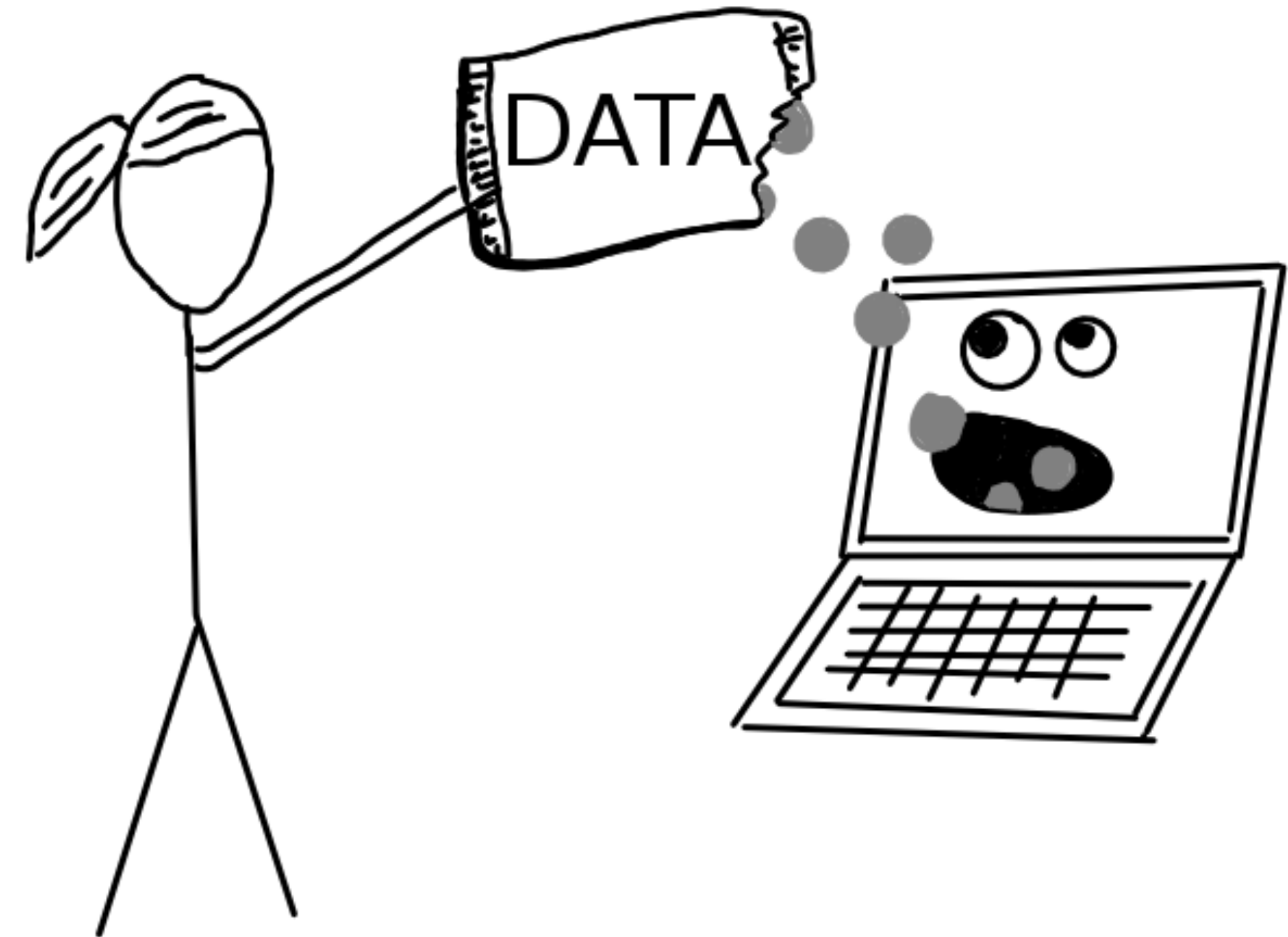
- Arthur Samuel (1959): Machine learning is the field of study that gives the computer the ability to learn **without being explicitly programmed**.



Without Machine Learning



With Machine Learning

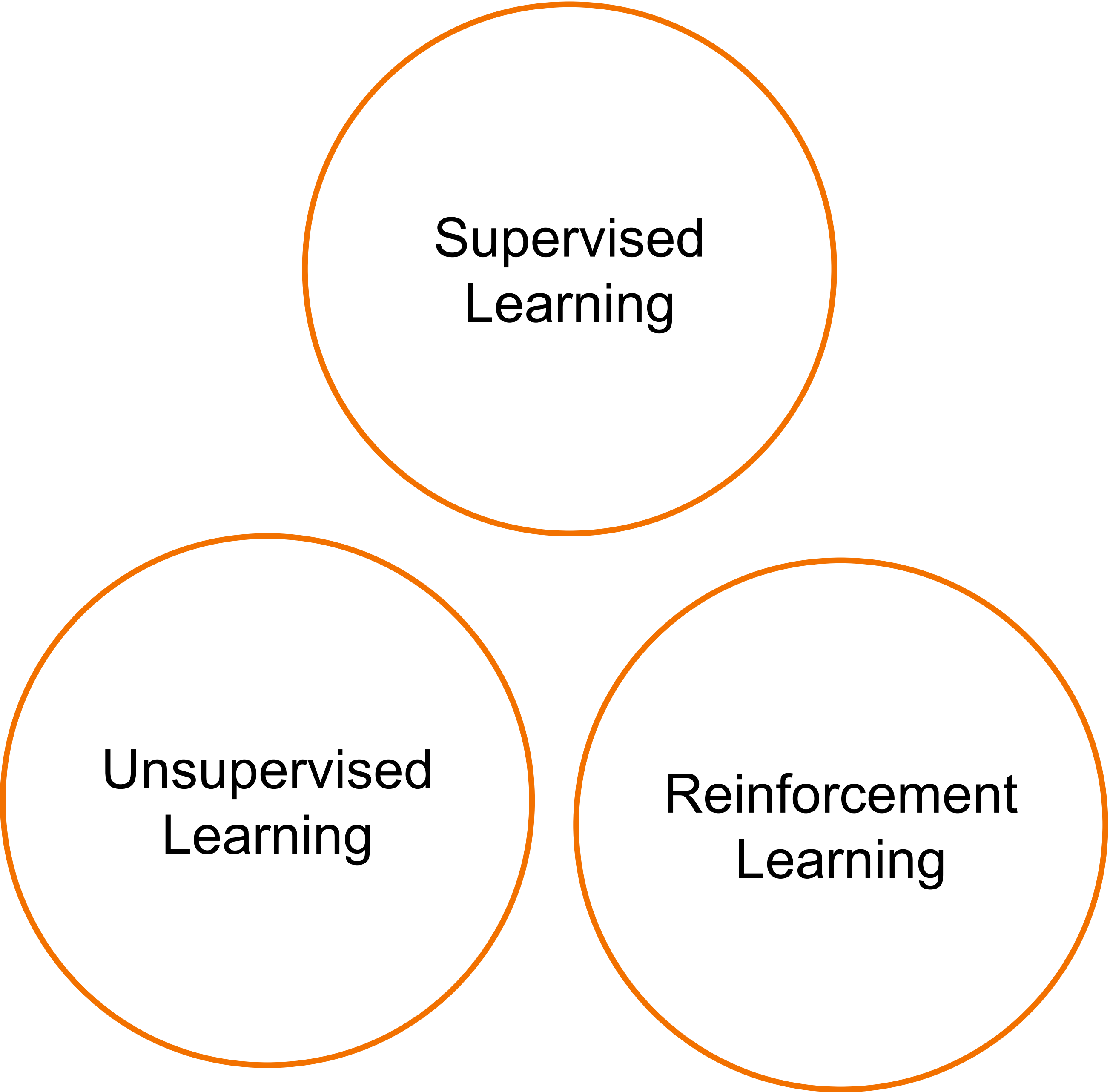


What is **machine learning**?

- Arthur Samuel (1959): Machine learning is the field of study that gives the computer the ability to learn **without being explicitly programmed**.
- Tom Mitchell (1997): A computer program is said to learn from **experience E** with respect to some class of **tasks T** and **performance measure P**, if its performance at tasks in T as measured by P, improves with experience E.



Taxonomy of ML



Supervised
Learning

Unsupervised
Learning

Reinforcement
Learning



Part II: Supervised Learning

Example 1: Predict whether a user likes a song or not



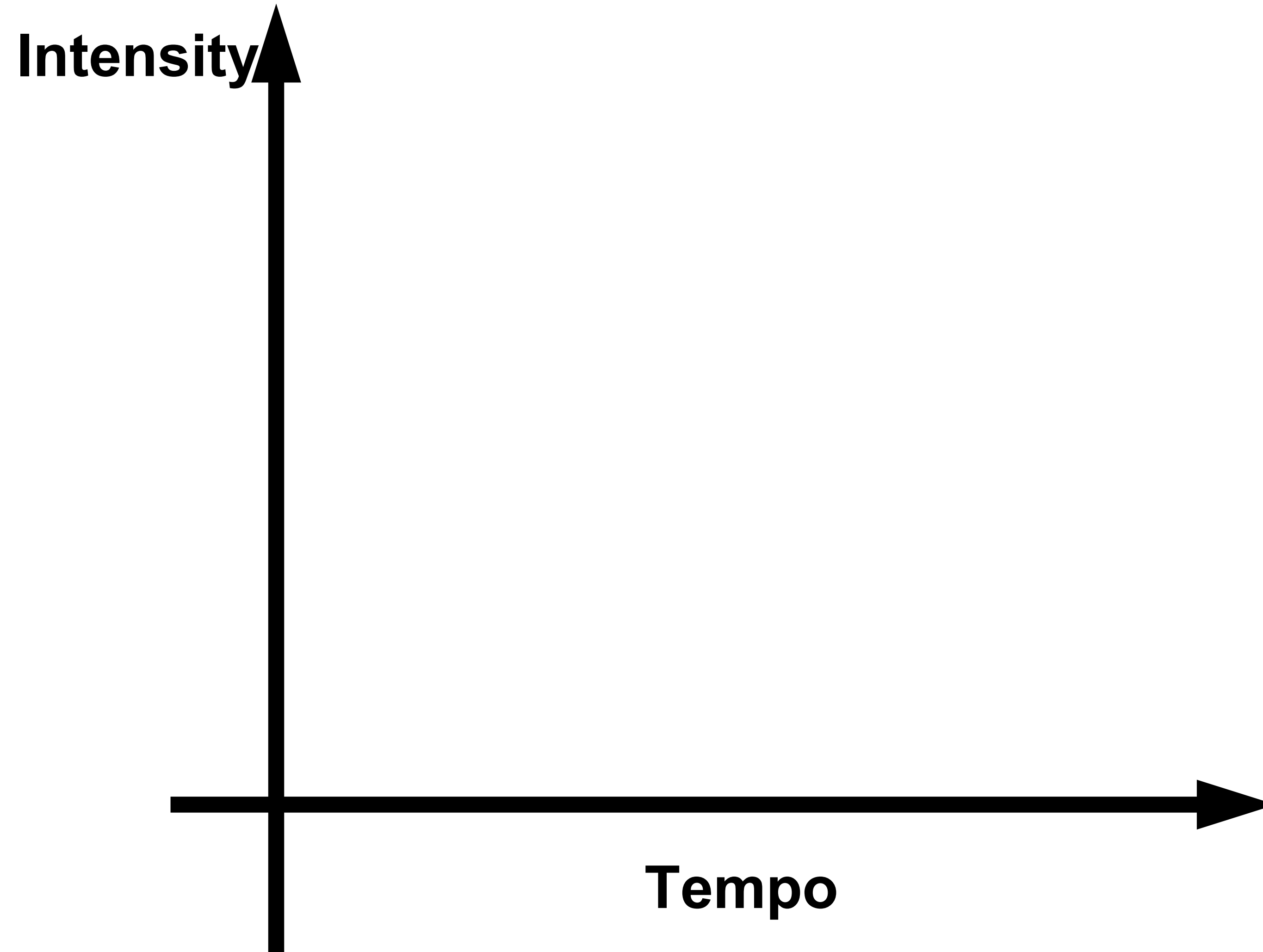
model



Example 1: Predict whether a user likes a song or not



User Sharon



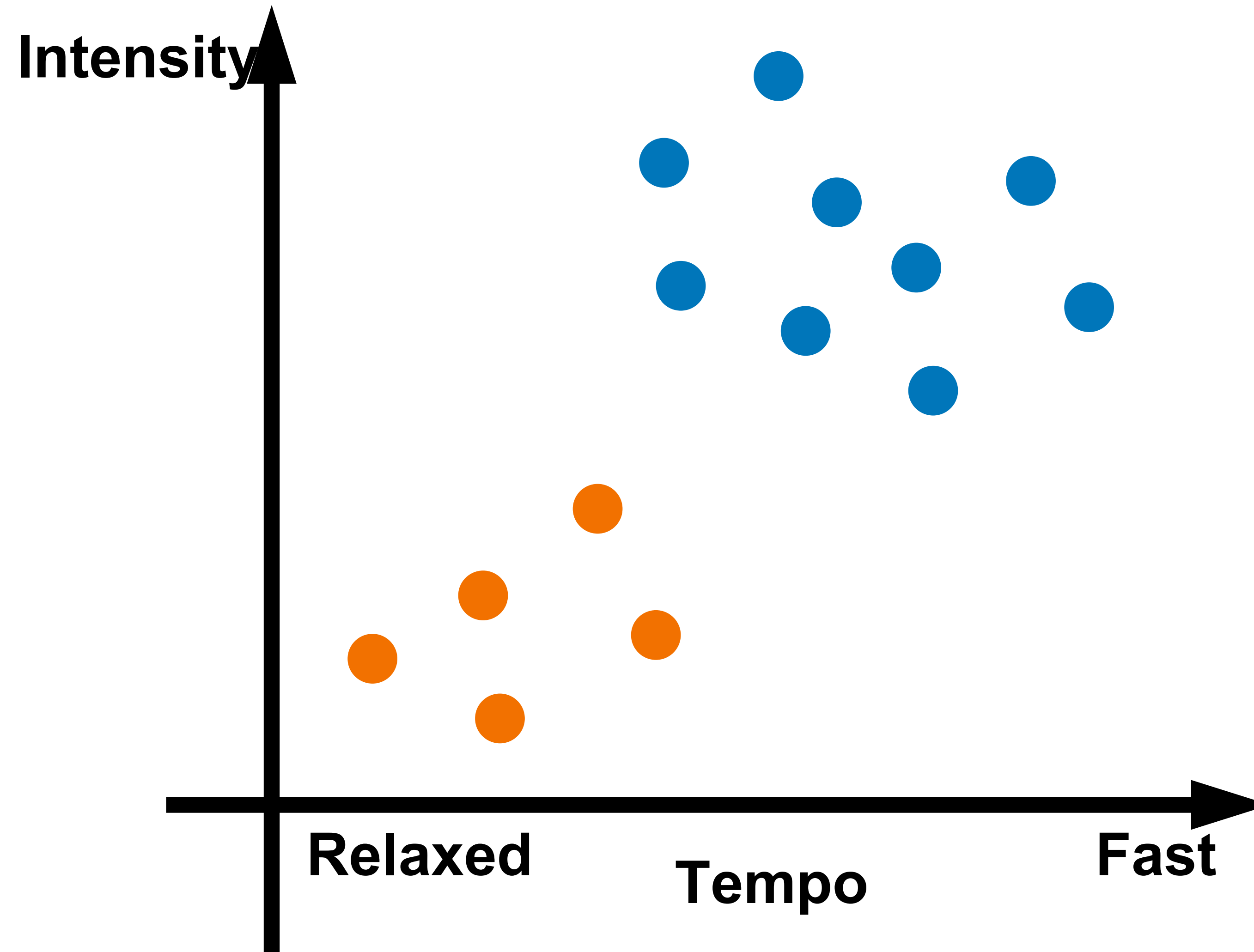
Example 1: Predict whether a user likes a song or not



User Sharon

● Dislike

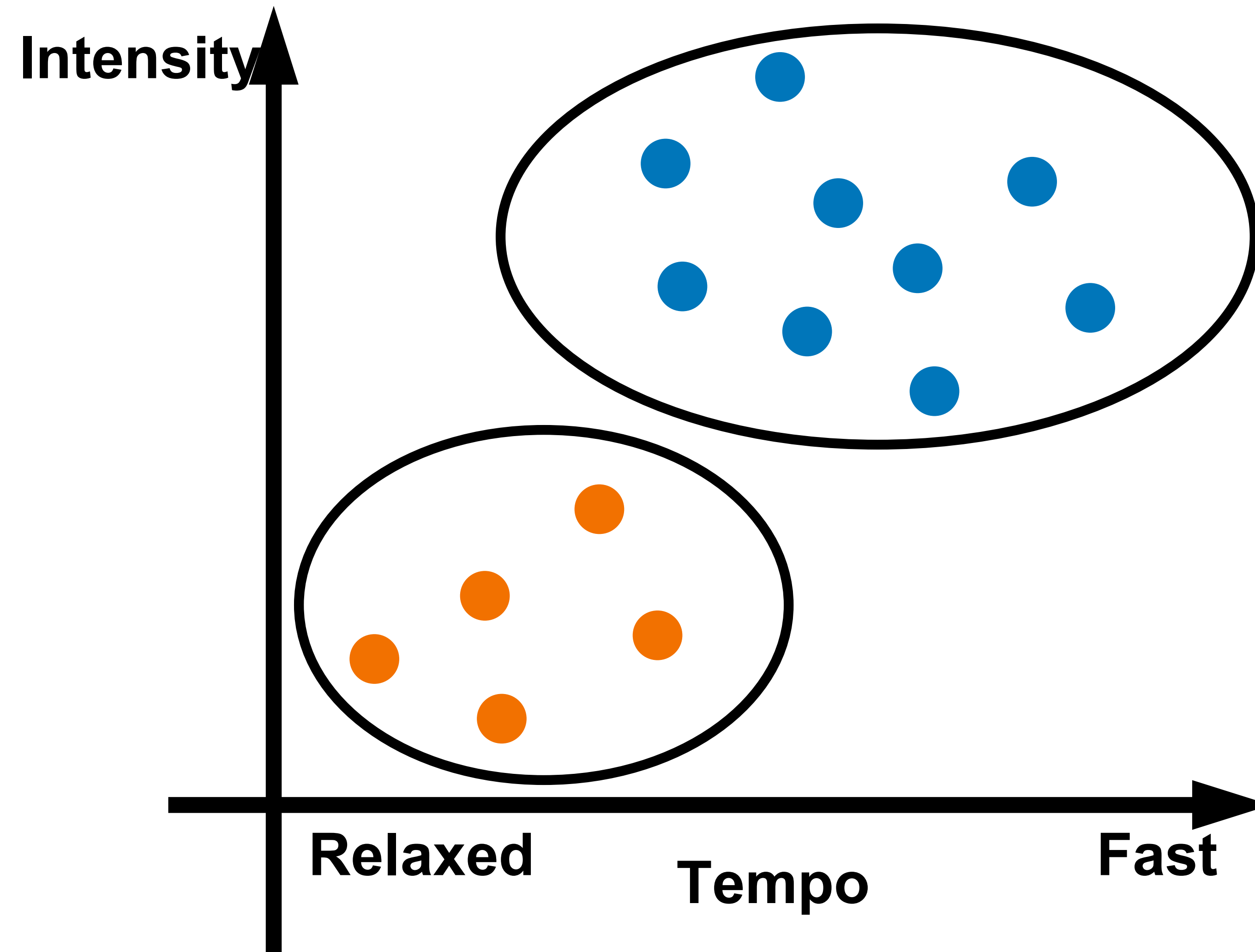
● Like



Example 1: Predict whether a user likes a song or not



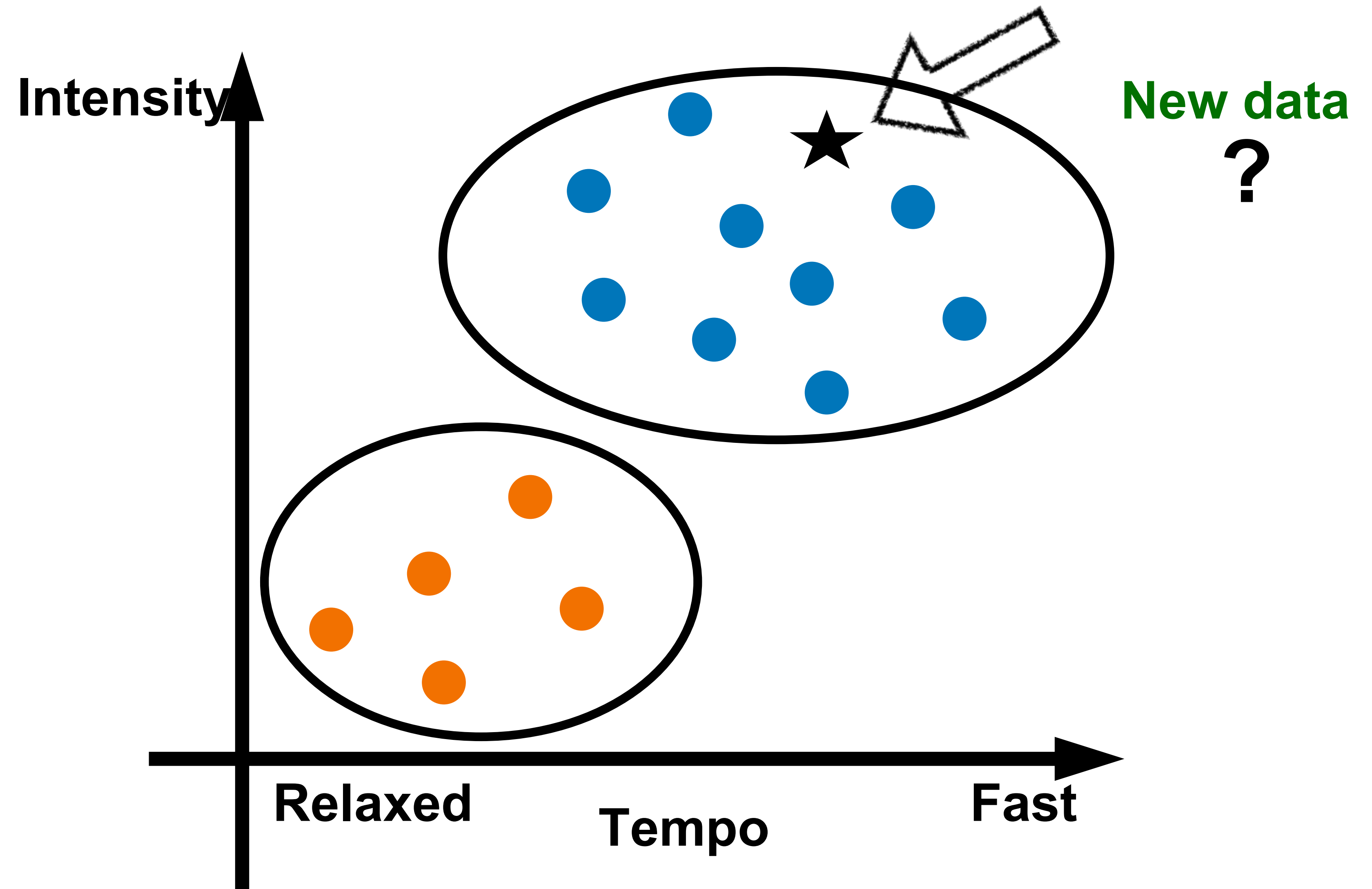
User Sharon



Example 1 : Predict whether a user likes a song or not



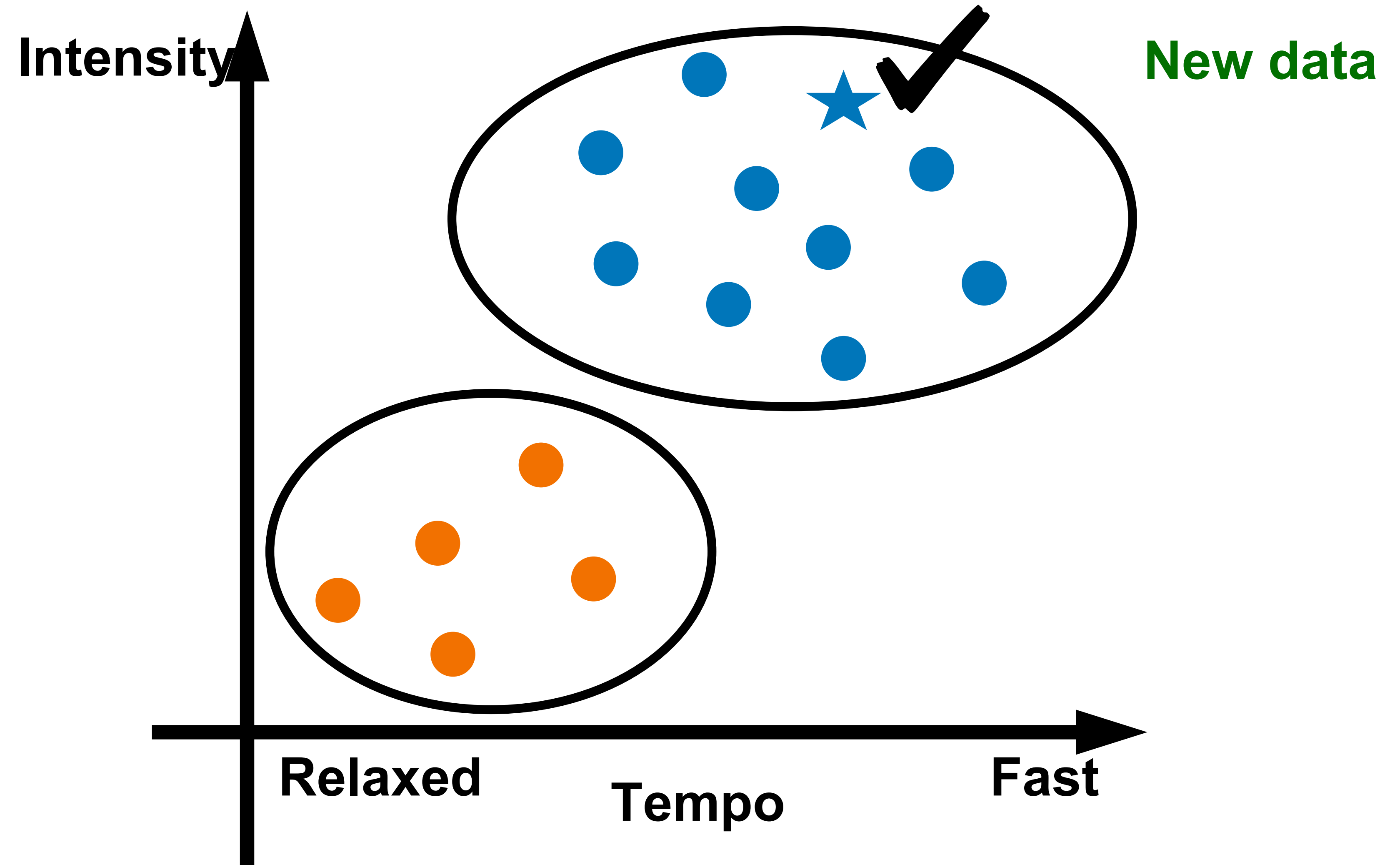
User Sharon



Example 1: Predict whether a user likes a song or not



User Sharon

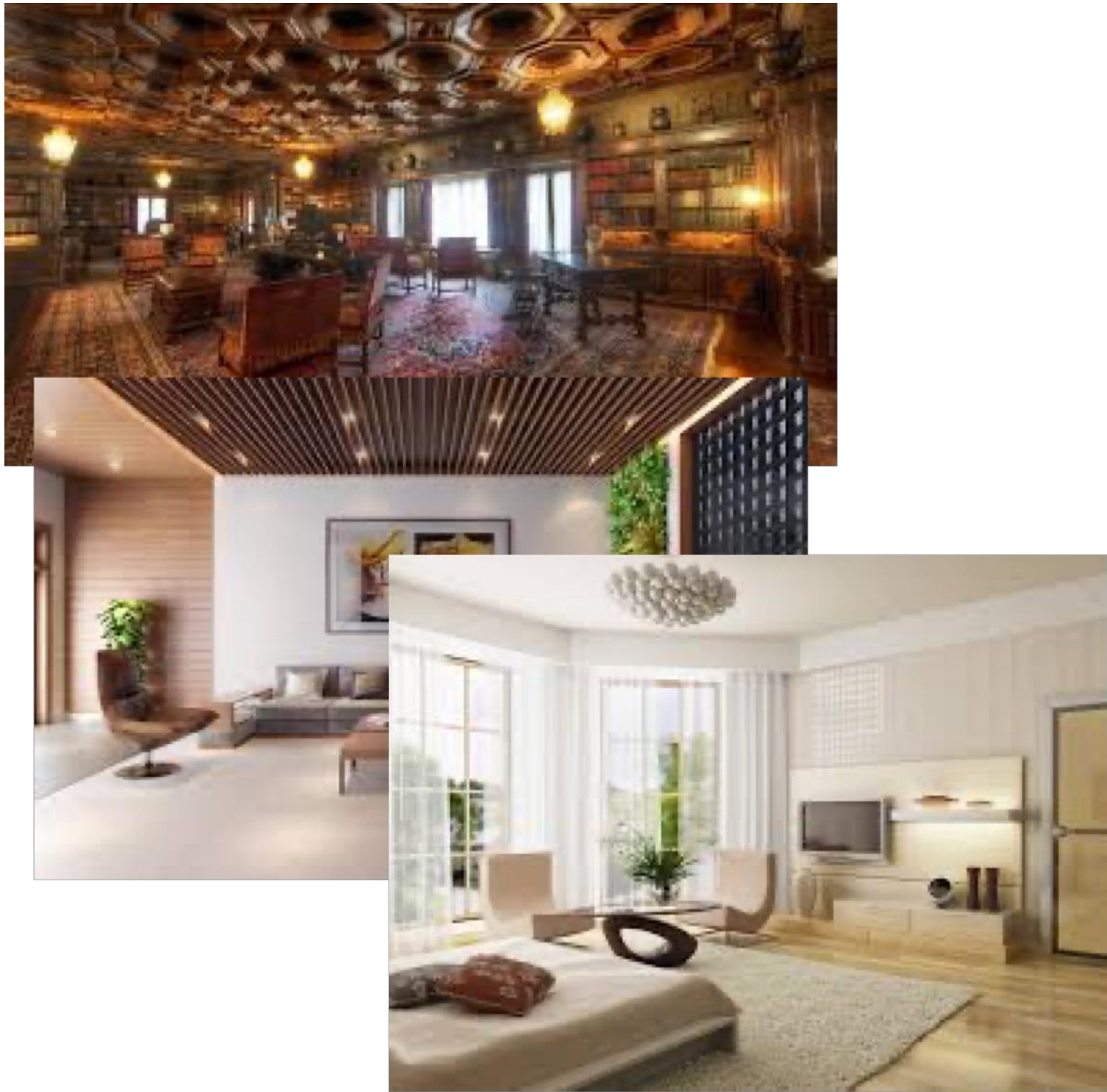


Example 2: Classify Images

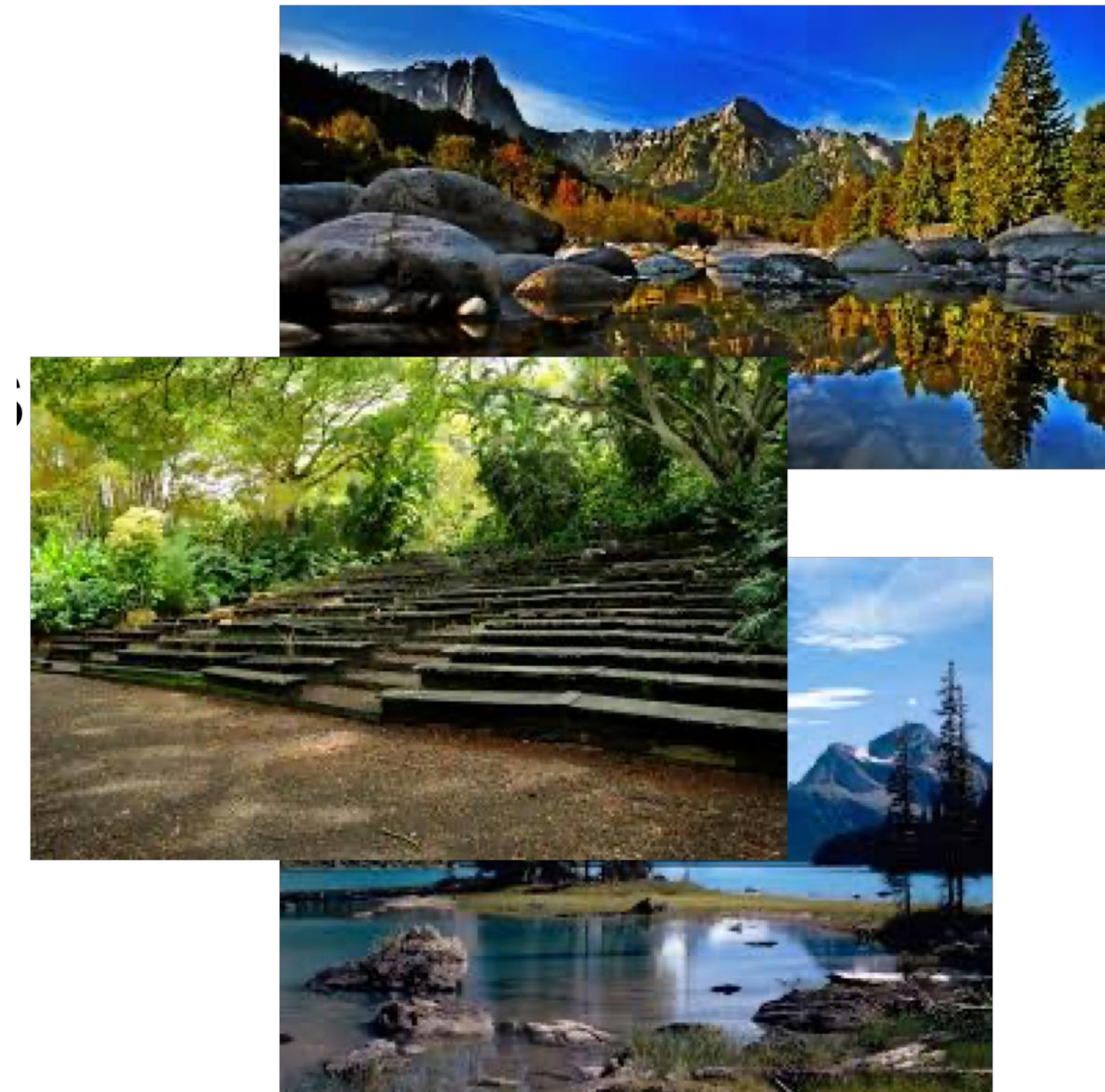
<http://www.image-net.org/>



Example 2: Classify Images

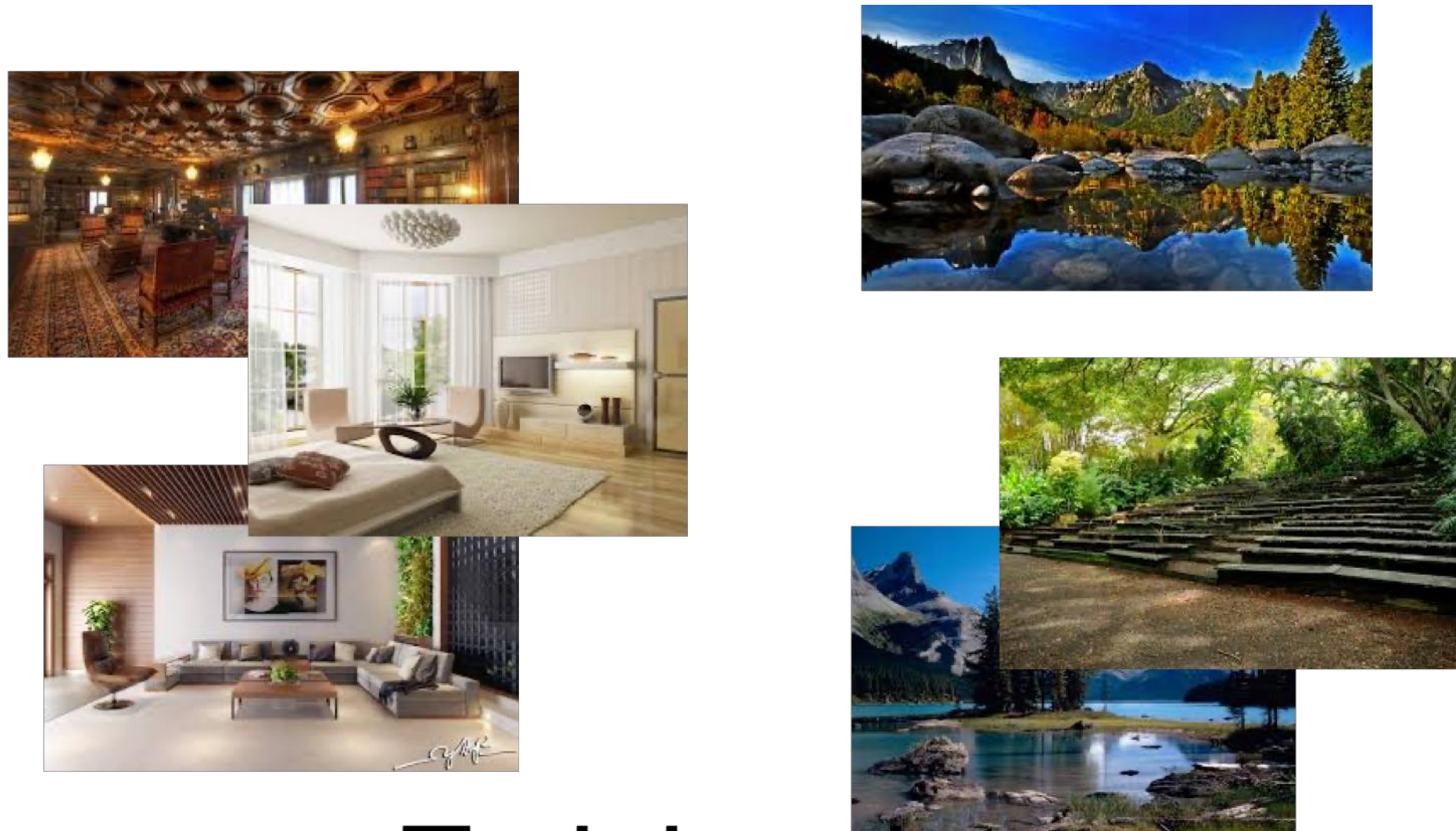


indoor



outdoor

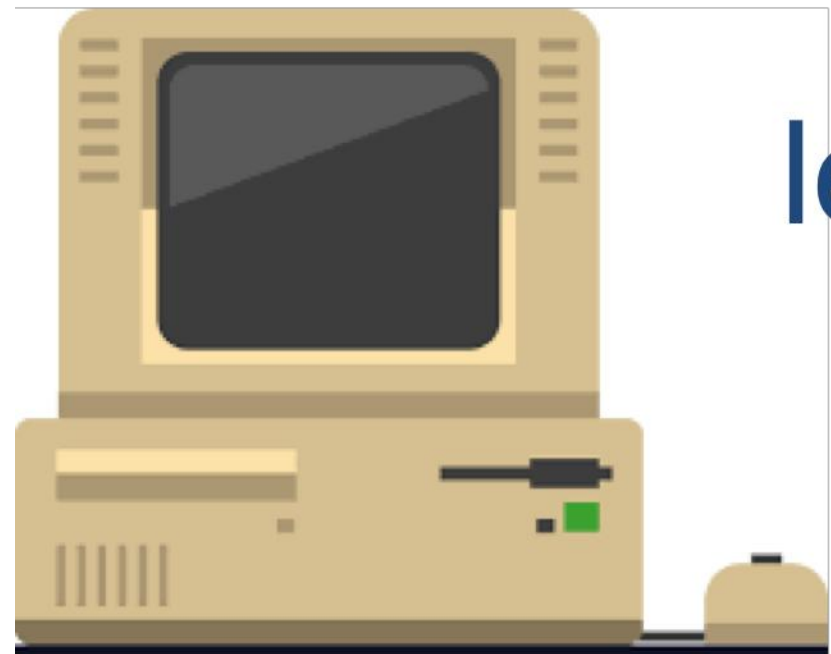
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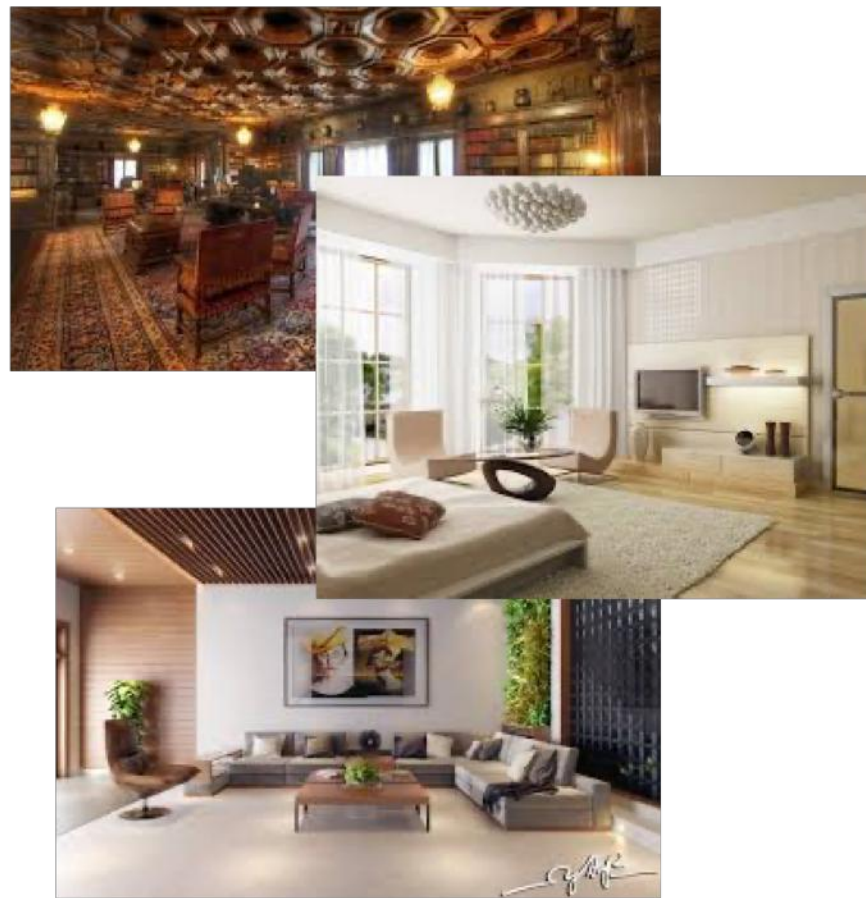


Training data



learning (i.e., training)





Label: outdoor



Label: indoor

Training data

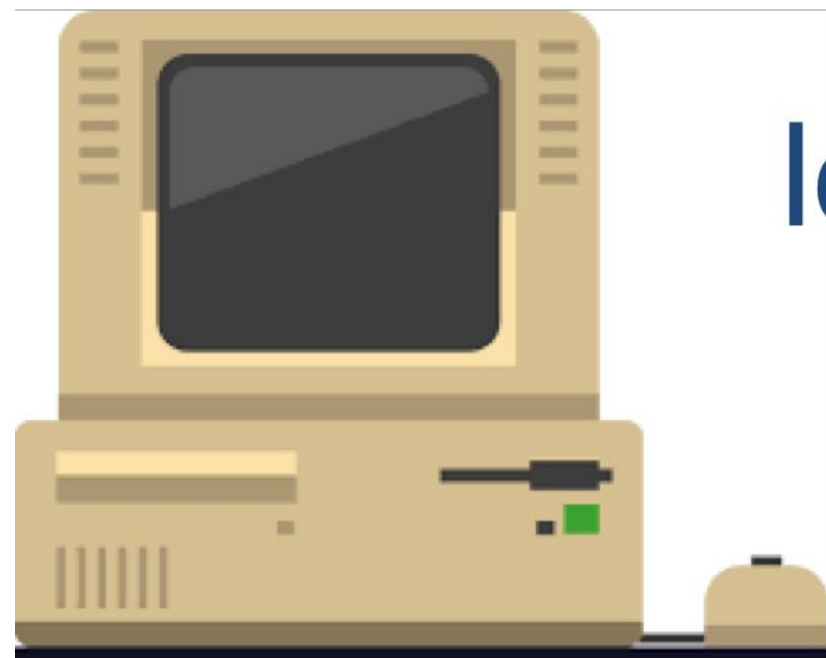
Test data



learning (i.e., training)

testing

performance



★ How to represent data?

input data

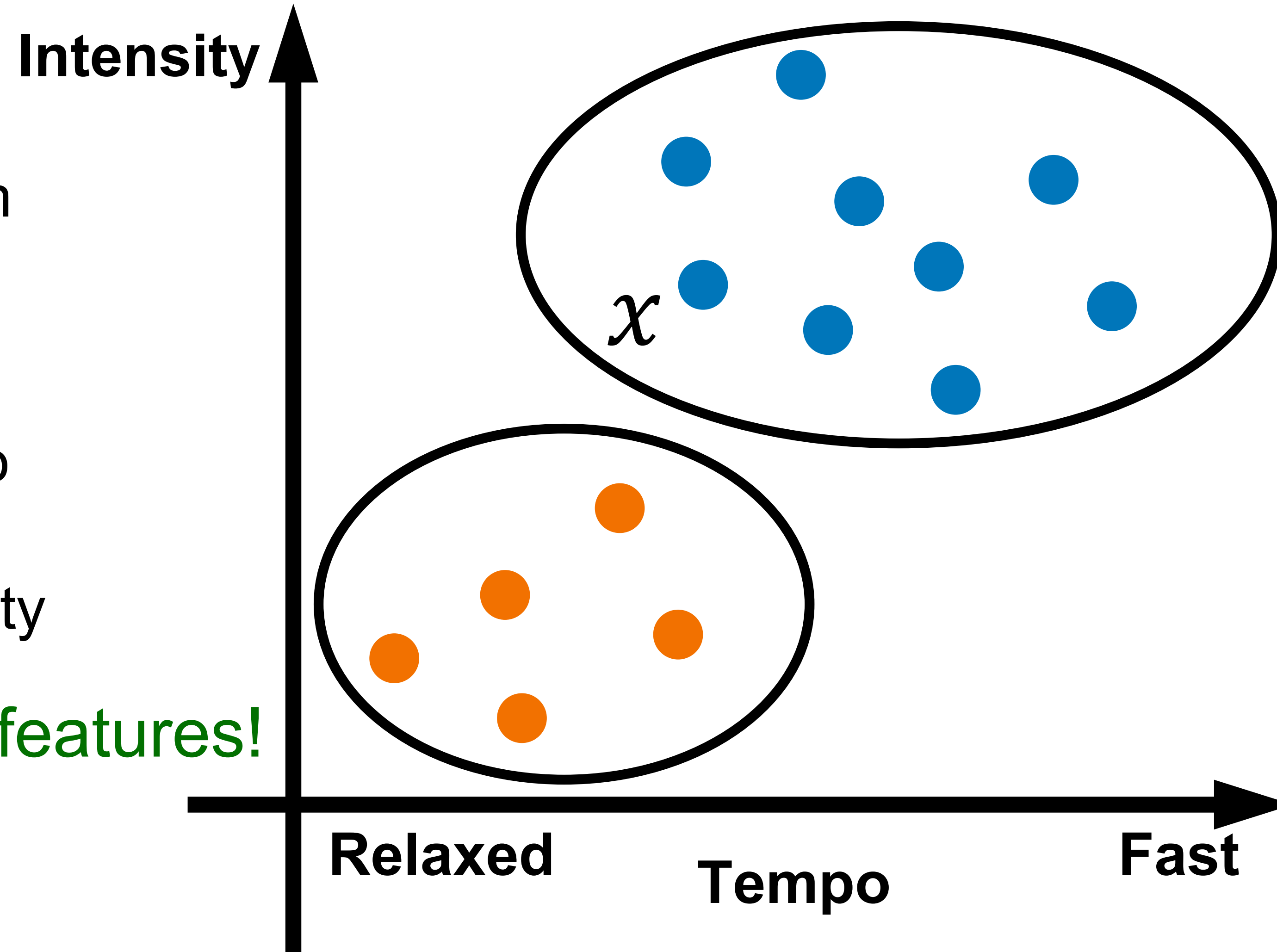
$$x \in \mathbb{R}^d$$

d : feature dimension

feature vector

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \begin{matrix} \text{Tempo} \\ \text{Intensity} \end{matrix}$$

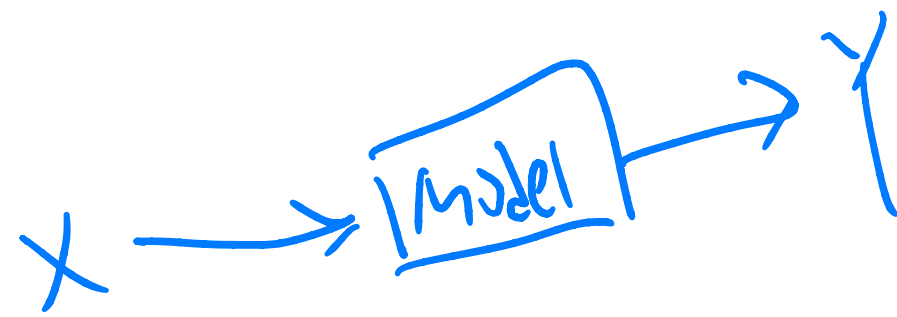
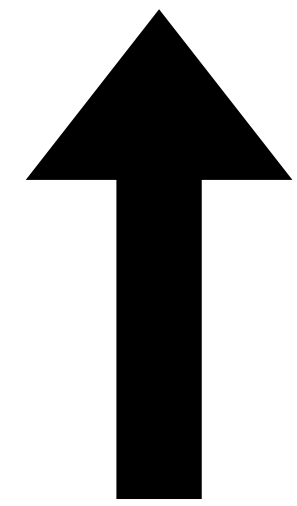
There can be many features!



How to represent data?

Label

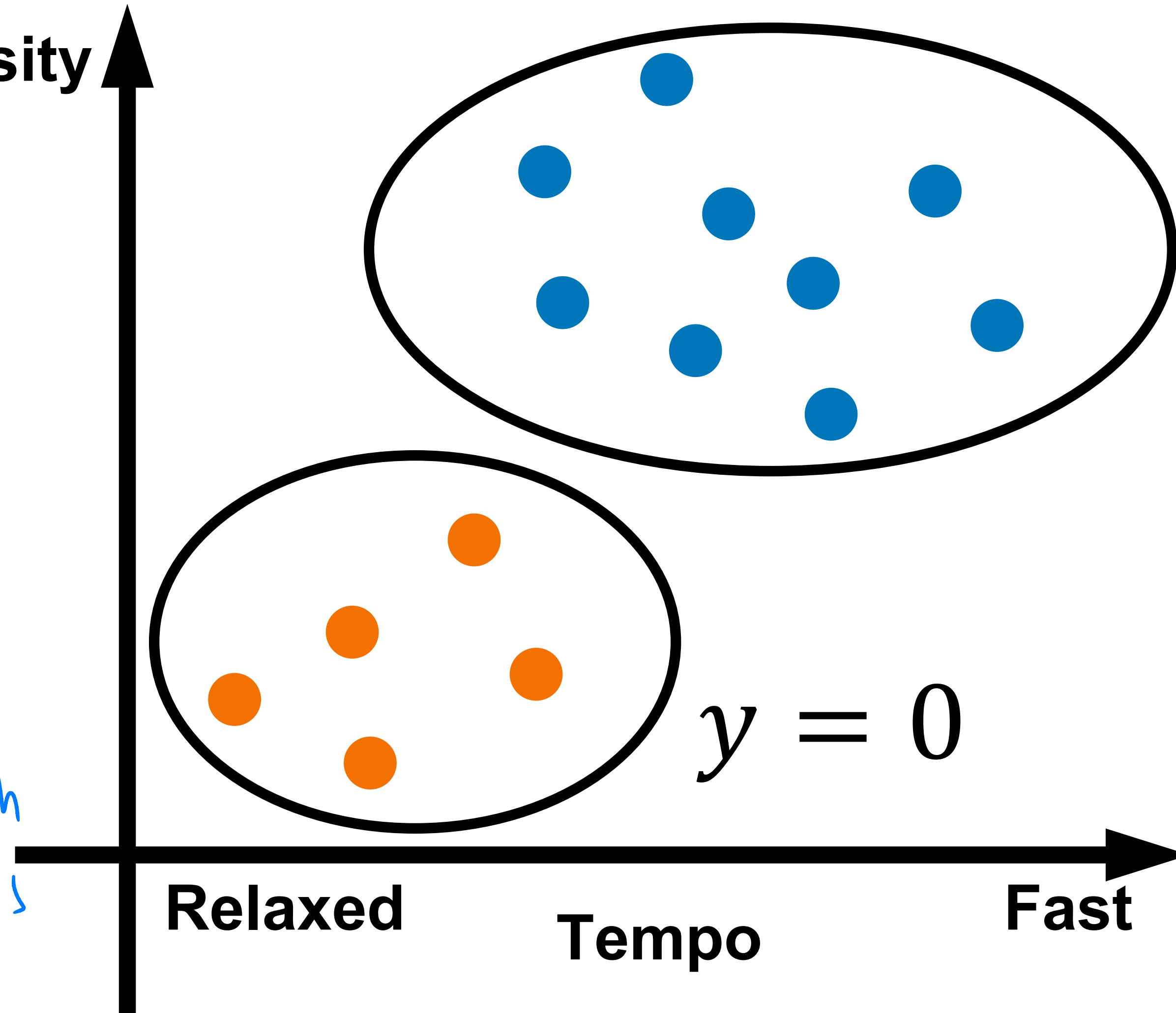
$$y \in \{0,1\}$$



★ Where “supervision”
comes from

my input: (x_1, y_1)
 (x_2, y_2)
 $\vdots$
 (x_n, y_n)
→ these are my
training data with
“correct” answers

Intensity



$$y = 1$$

$$y = 0$$

Relaxed

Tempo

Fast

Represent various types of data

- Image
 - Pixel values
- Bank account
 - Credit rating, balance, # deposits in last day, week, month, year, #withdrawals

Two Types of Supervised Learning Algorithms

Classification



Discrete outputs

Regression



Continuous values

Example of regression: housing price prediction

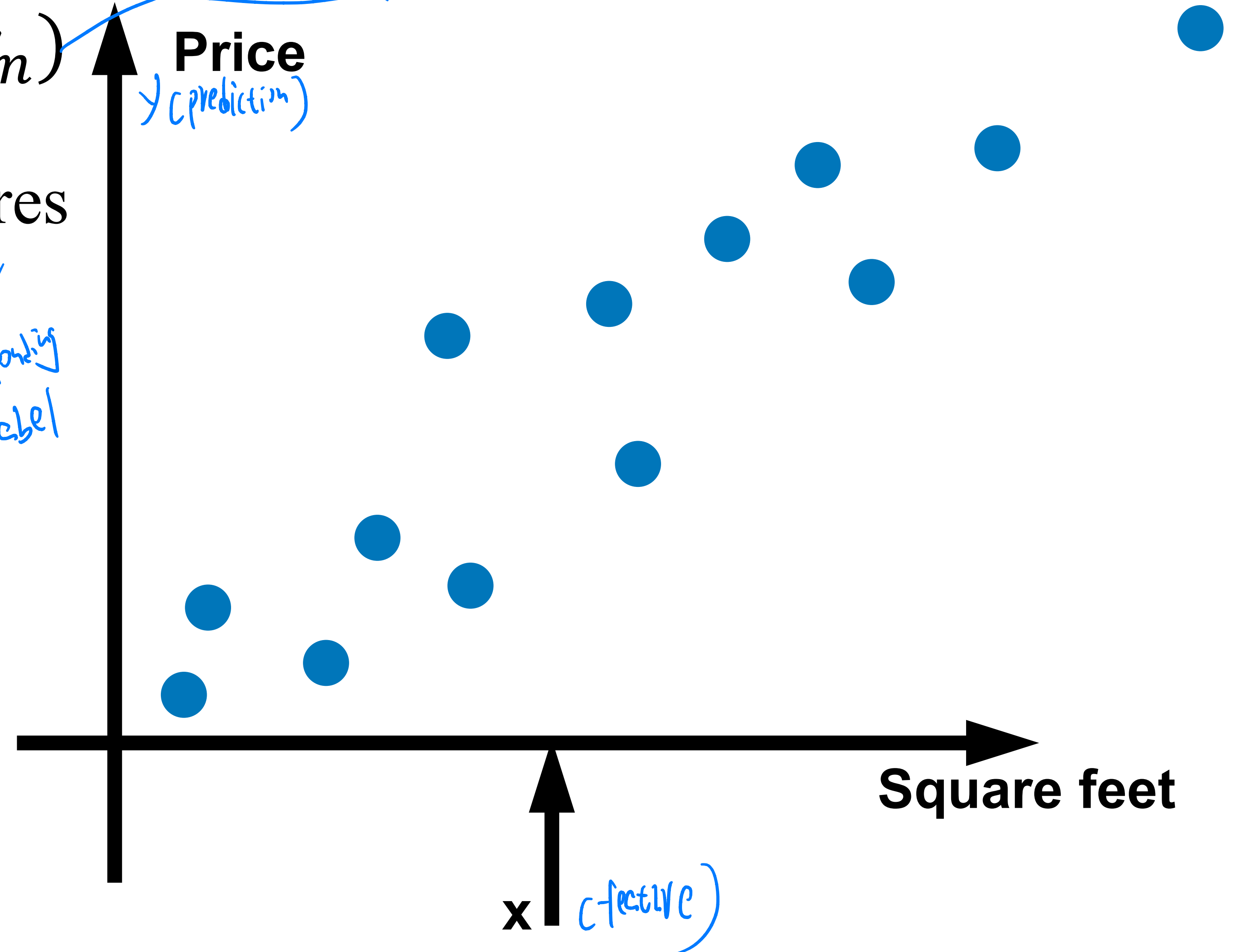
Given: a dataset that contains n samples

$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$

Task: if a residence has x squares feet, predict the price?

In this case, $x \in \mathbb{R}^1$

feature
corresponding label



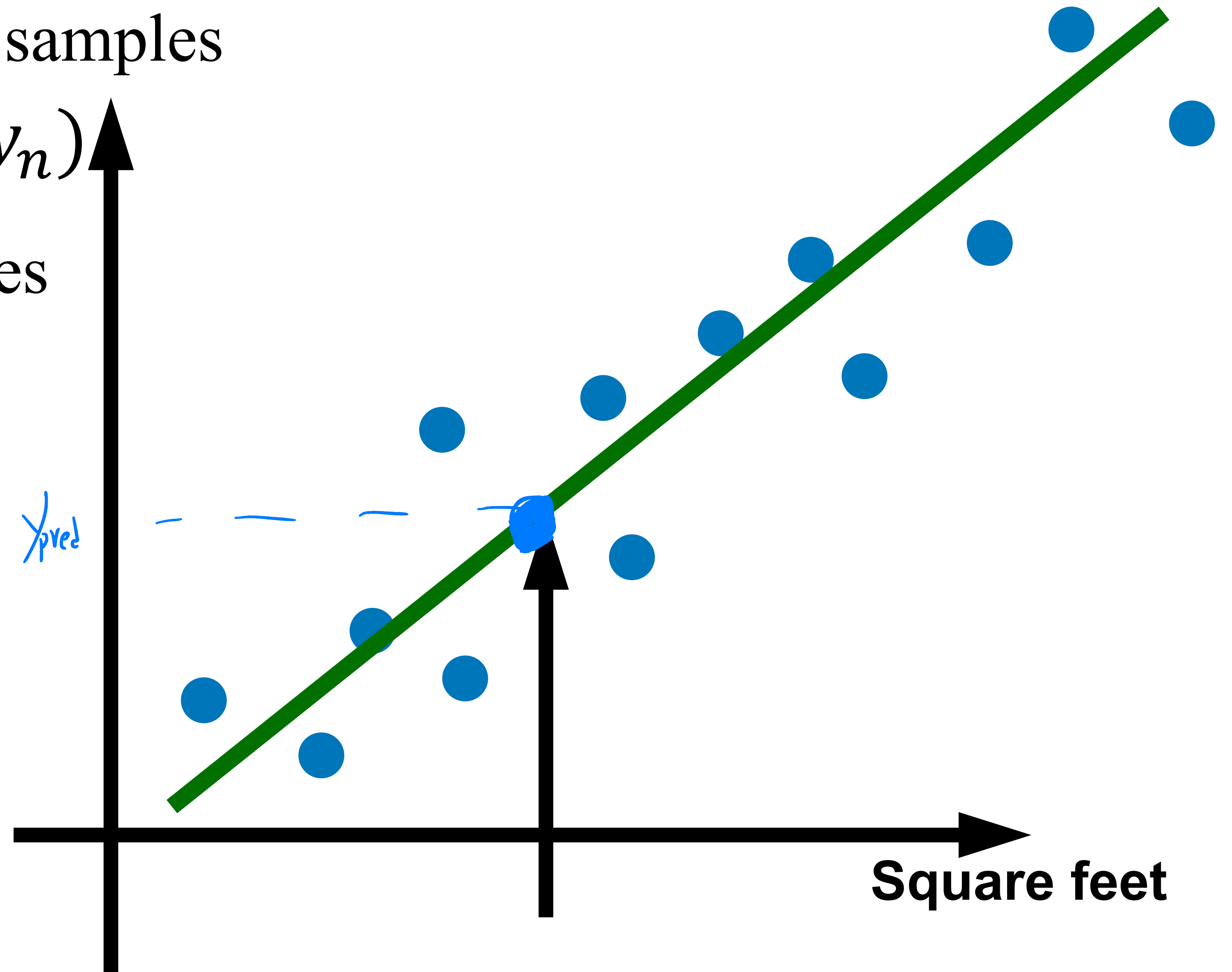
Example of regression: housing price prediction

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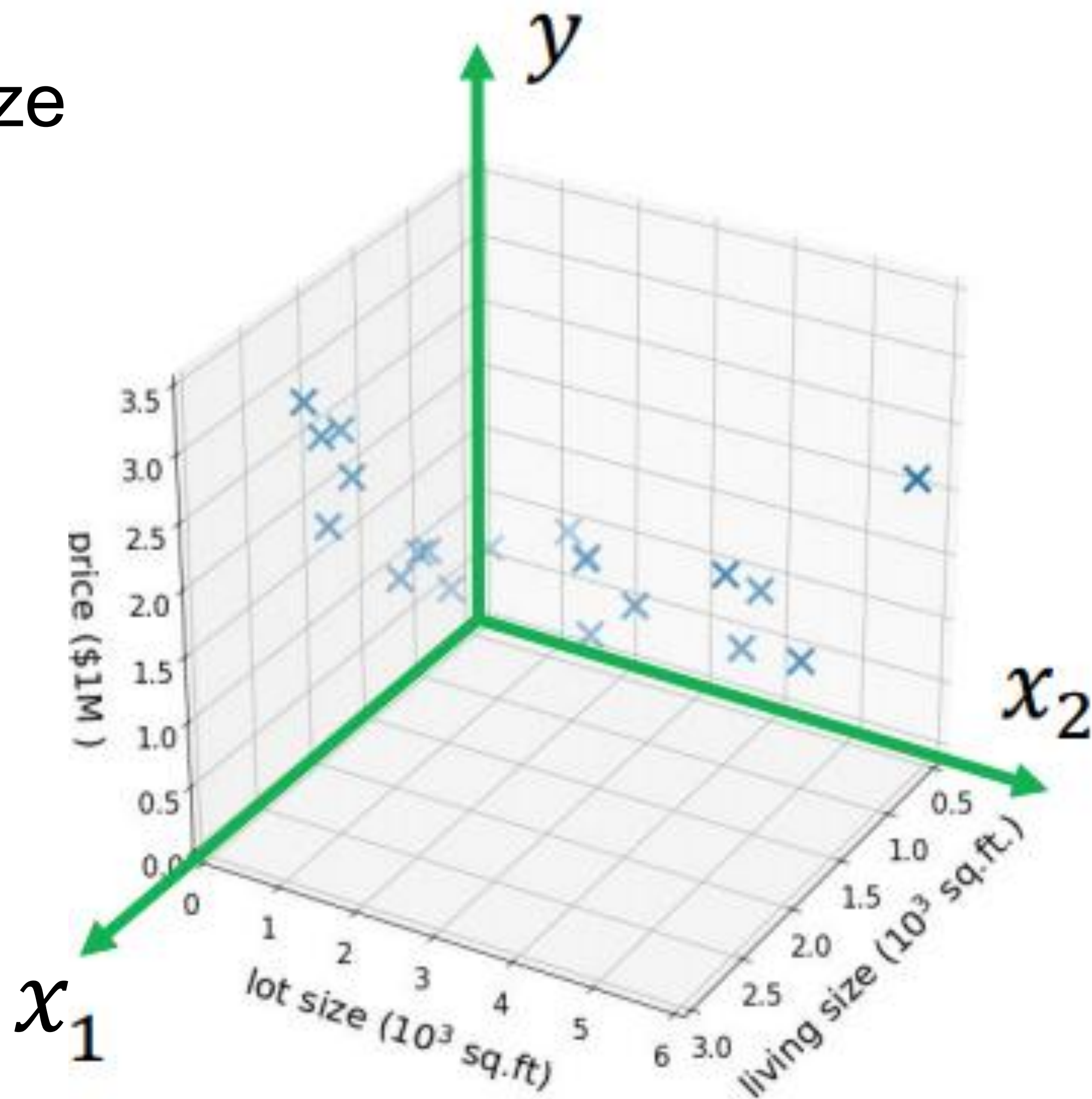
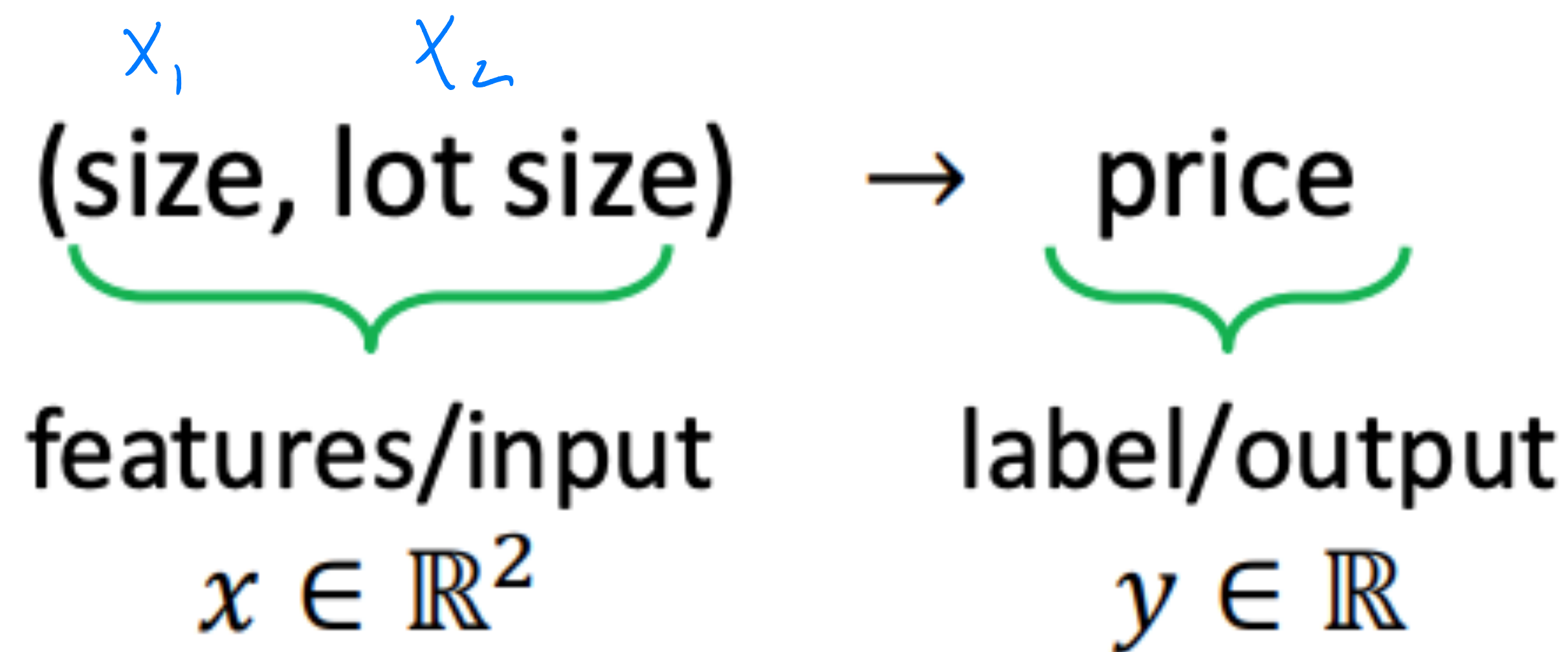
Task: if a residence has x squares feet, predict the price?

$$y \in \mathbb{R}$$



Example of regression: housing price prediction

Input with more features (e.g., lot size

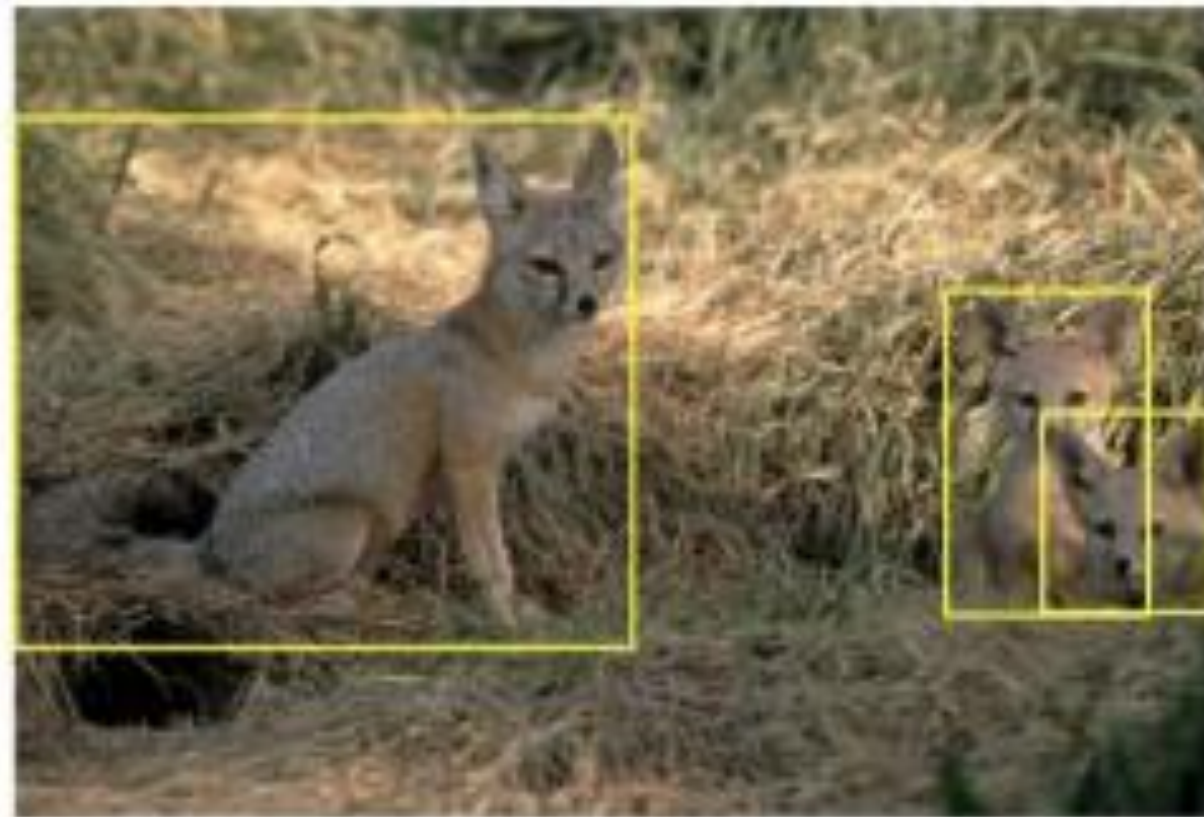


(credit: stanford CS229)

Supervised Learning: More examples

x = raw pixels of the image

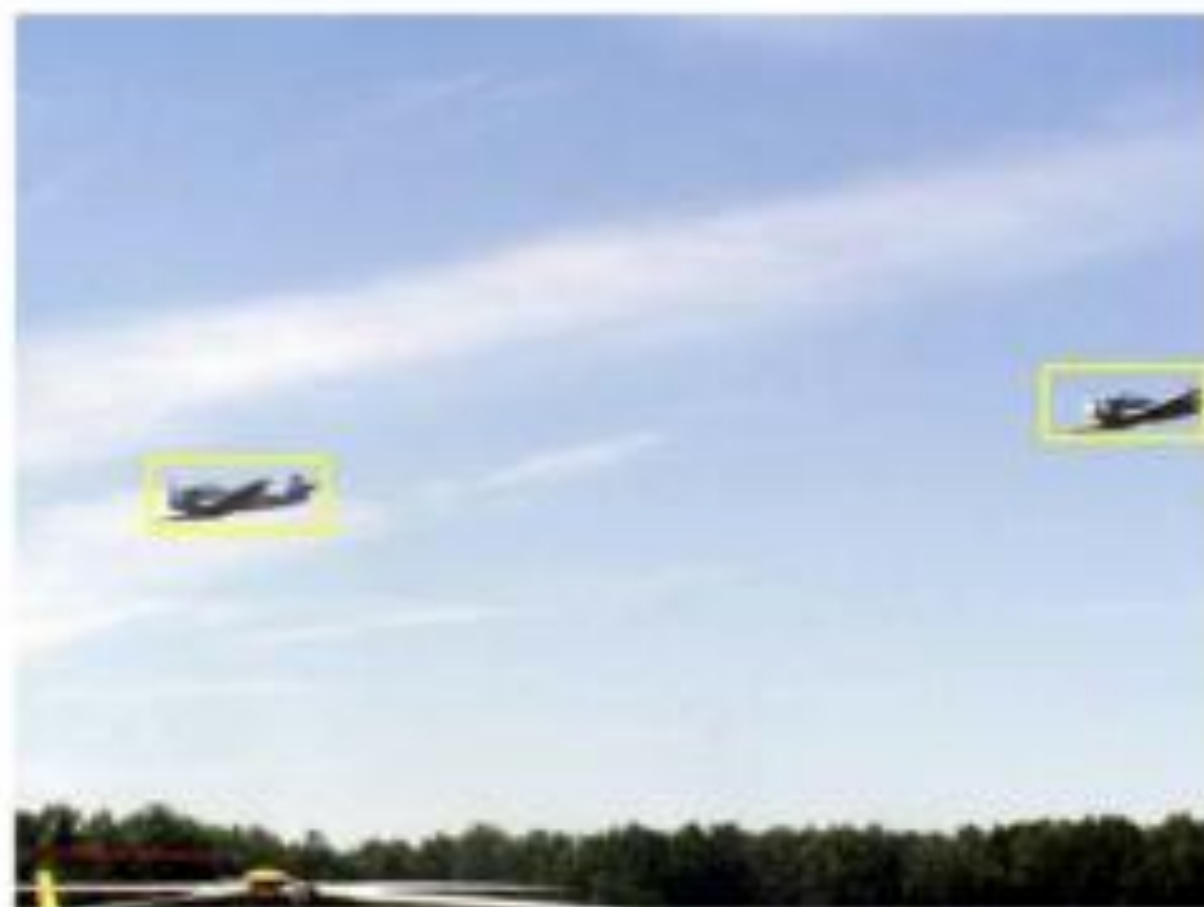
y = bounding boxes



kit fox



croquette



airplane



frog

★ Two Types of Supervised Learning Algorithms

Classification

- the label is a **discrete** variable

$$y \in \{1, 2, 3, \dots, K\}$$

Regression

- the label is a **continuous** variable

$$y \in \mathbb{R}$$

Training Data for Supervised Learning

Training data is a collection of input instances to the learning algorithm:

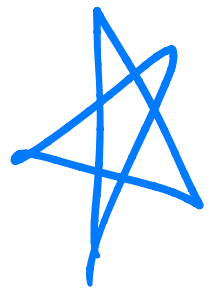
$$(\underbrace{x_1}_{\text{input}}, \underbrace{y_1}_{\text{label}}), (x_2, y_2), (x_3, y_3), \dots, (x_n, y_n)$$

★ The training data is the “**experience**” given to a learning algorithm

Goal of Supervised Learning

Given training data

$$(x_1, y_1), (x_2, y_2), (x_3, y_3), \dots, (x_n, y_n)$$



Learn a function mapping $f: X \rightarrow Y$, such that $f(x)$
predicts the label y on **future** data x (not in training data)

Goal of Supervised Learning

Training set error

- 0-1 loss for classification $\ell = \frac{1}{n} \sum_{i=1}^n (f(\mathbf{x}_i) \neq y_i)$
Handwritten notes: y_i (with an arrow pointing to the y_i in the formula), 0 indicates perfect, 1 indicates all wrong
- Squared loss for regression: $\ell = \frac{1}{n} \sum_{i=1}^n (f(\mathbf{x}_i) - y_i)^2$
Handwritten note: (How far we are off?)

A learning algorithm optimizes the training objective

$$f^* = \operatorname{argmin} \mathbb{E}_{(x,y)} \ell(f(x), y)$$

Handwritten note: minimize the loss function

Details in upcoming lectures :)

Quiz Break

Q1-1: Which is true about feature vectors?

- A. Feature vectors can have at most 10 dimensions
- B. Feature vectors have only numeric values
- C. The raw image can also be used as the feature vector
- D. Text data don't have feature vectors

Quiz Break

Q1-1: Which is true about feature vectors?

- A. Feature vectors can have at most 10 dimensions
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- D. Text data don't have feature vectors

- A. Feature vectors can be high dimensional
- B. Some feature vectors can have other types of values like strings
- D. Bag-of-words is a type of feature vector for text

Quiz Break

Q1-2: Which of the following is not a common task of supervised learning?

- A. Object detection (predicting bounding box from raw images)
- B. Classification
- C. Regression
- D. Dimensionality reduction

Quiz Break

Q1-2: Which of the following is not a common task of supervised learning?

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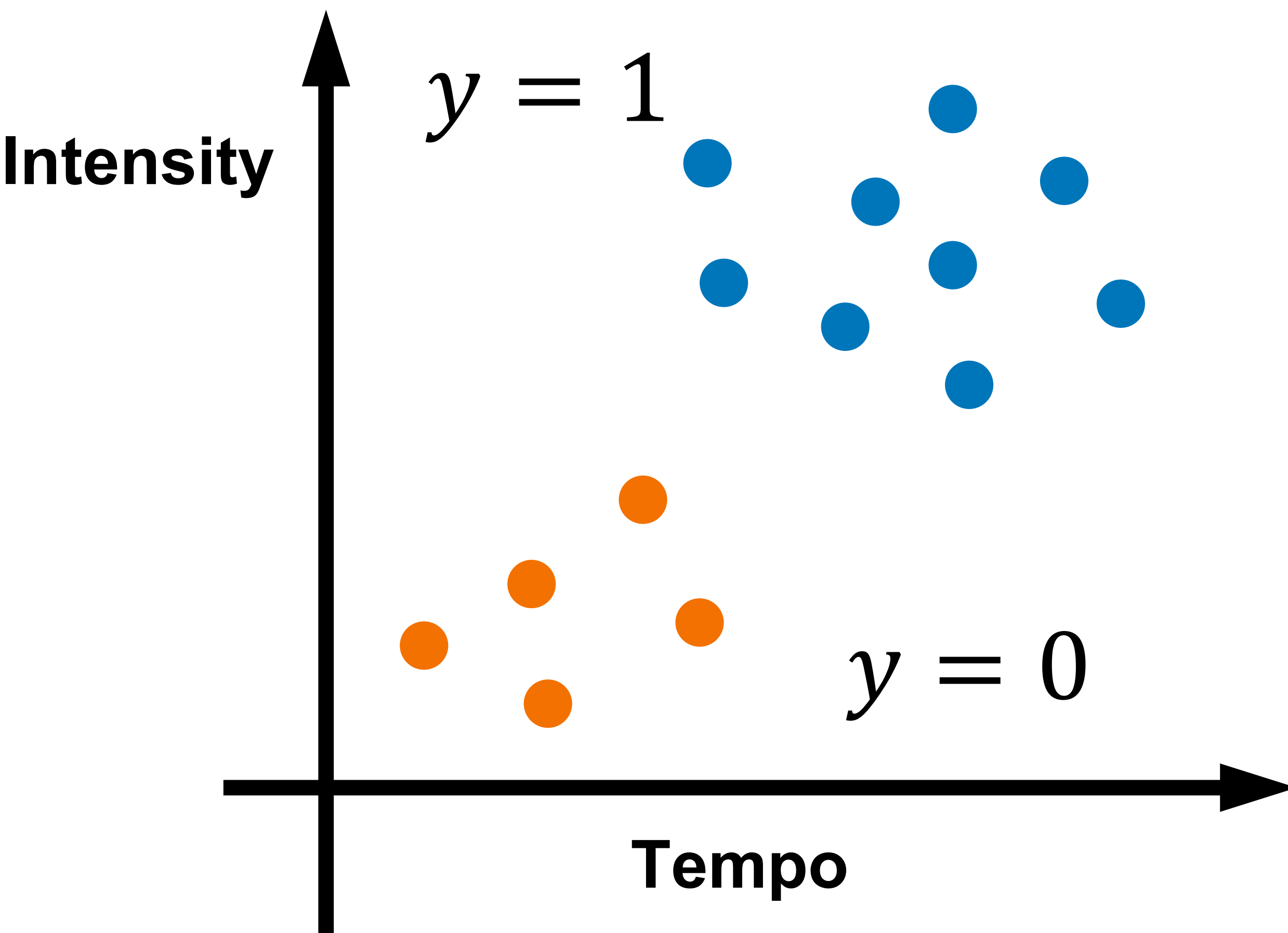
Part II: Unsupervised Learning (no teacher)

Unsupervised Learning

- Given: dataset contains **no label** $x_1, x_2, \dots, x_n$
- **Goal:** discover interesting patterns and structures in the data

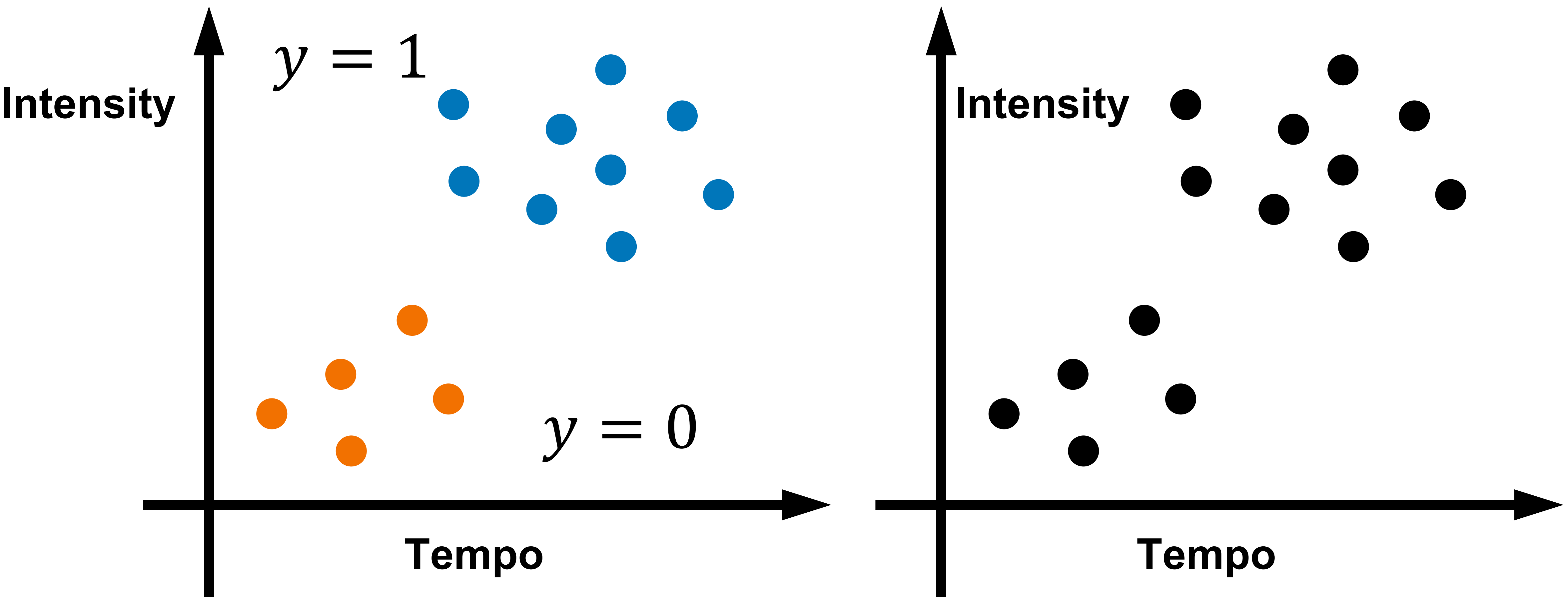
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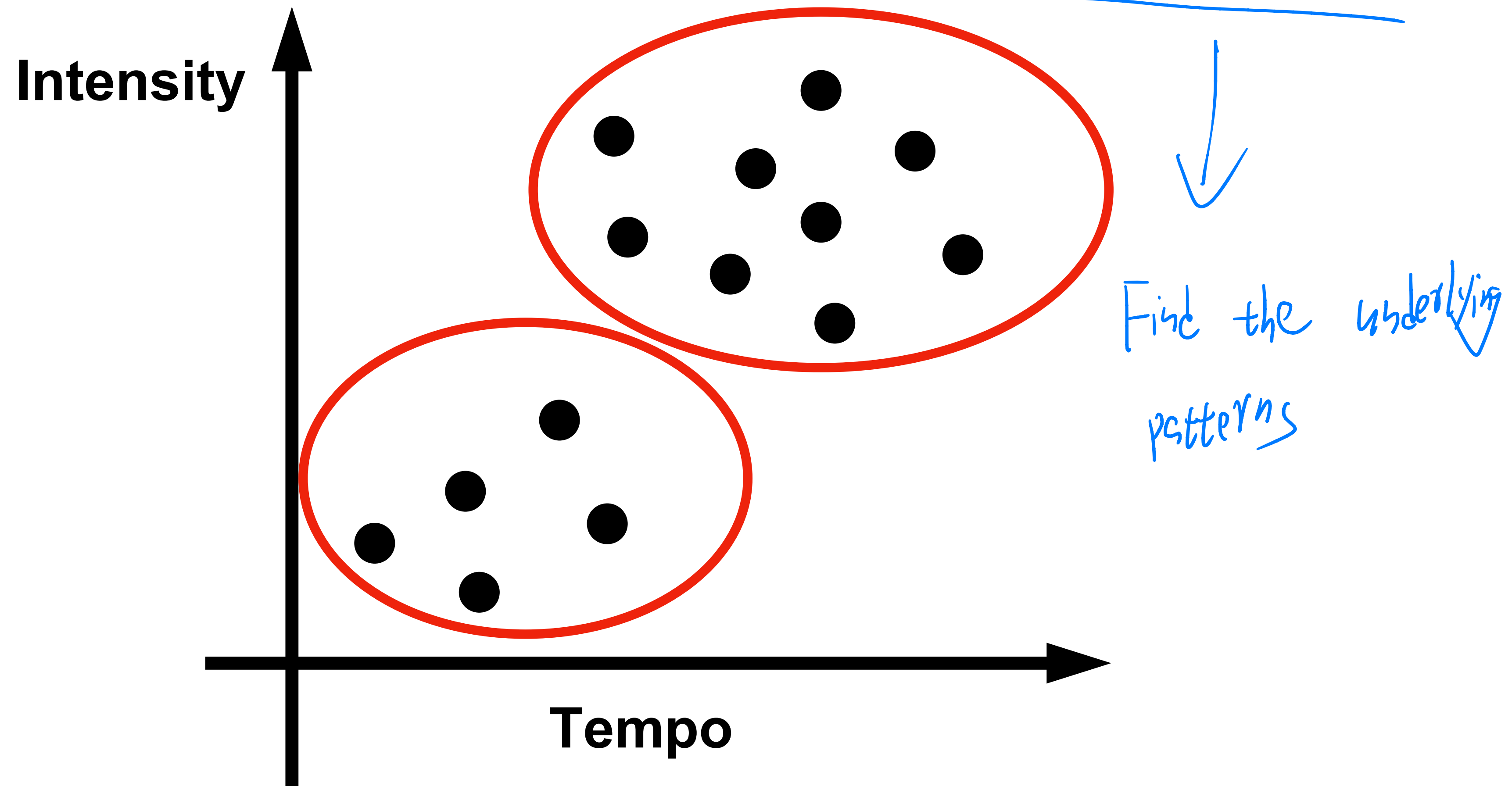
Unsupervised Learning

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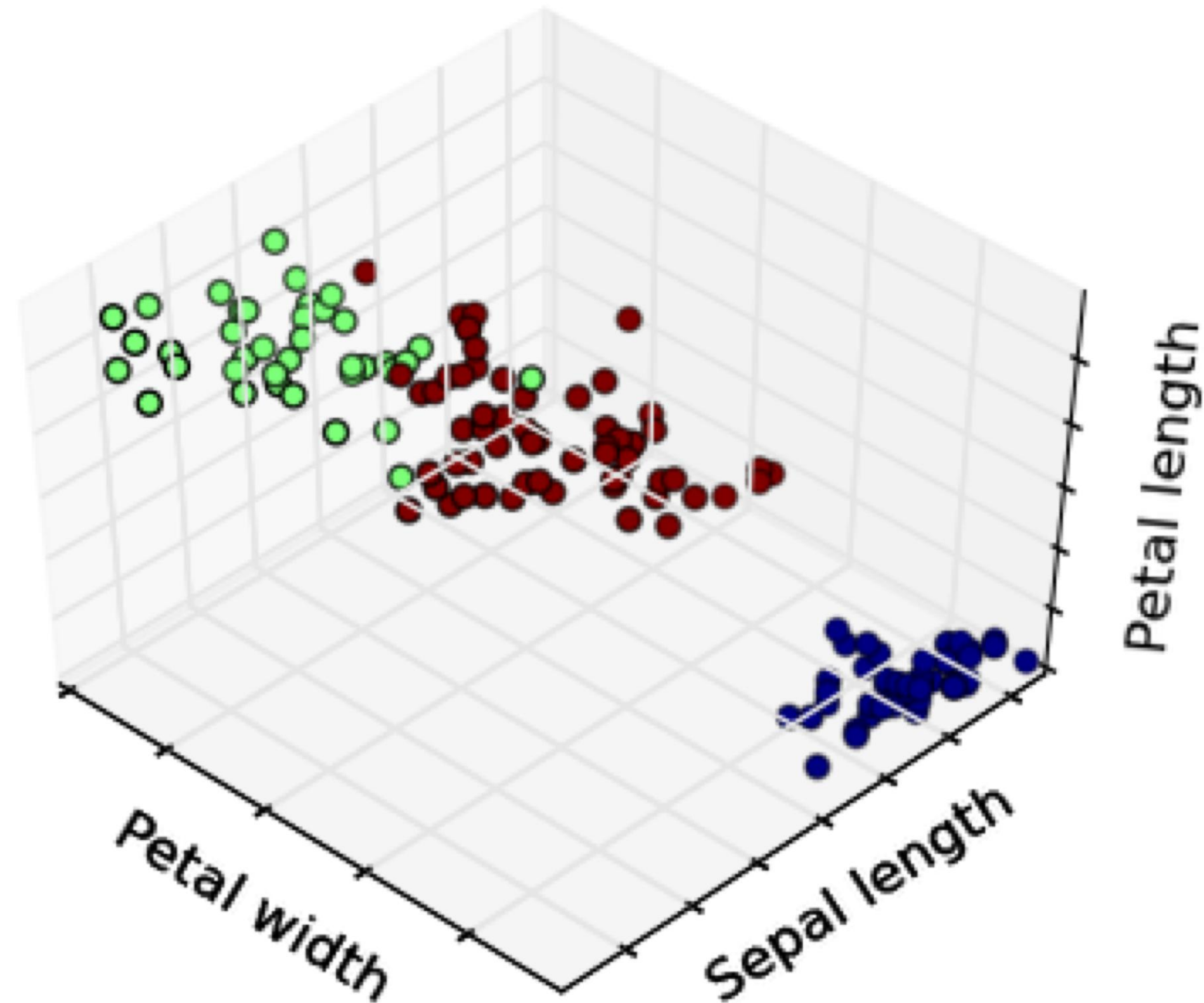


Clustering

- Given: dataset contains **no label** $x_1, x_2, \dots, x_n$
- **Output:** divides the data into clusters such that there are intra-cluster similarity and inter-cluster dissimilarity



Clustering

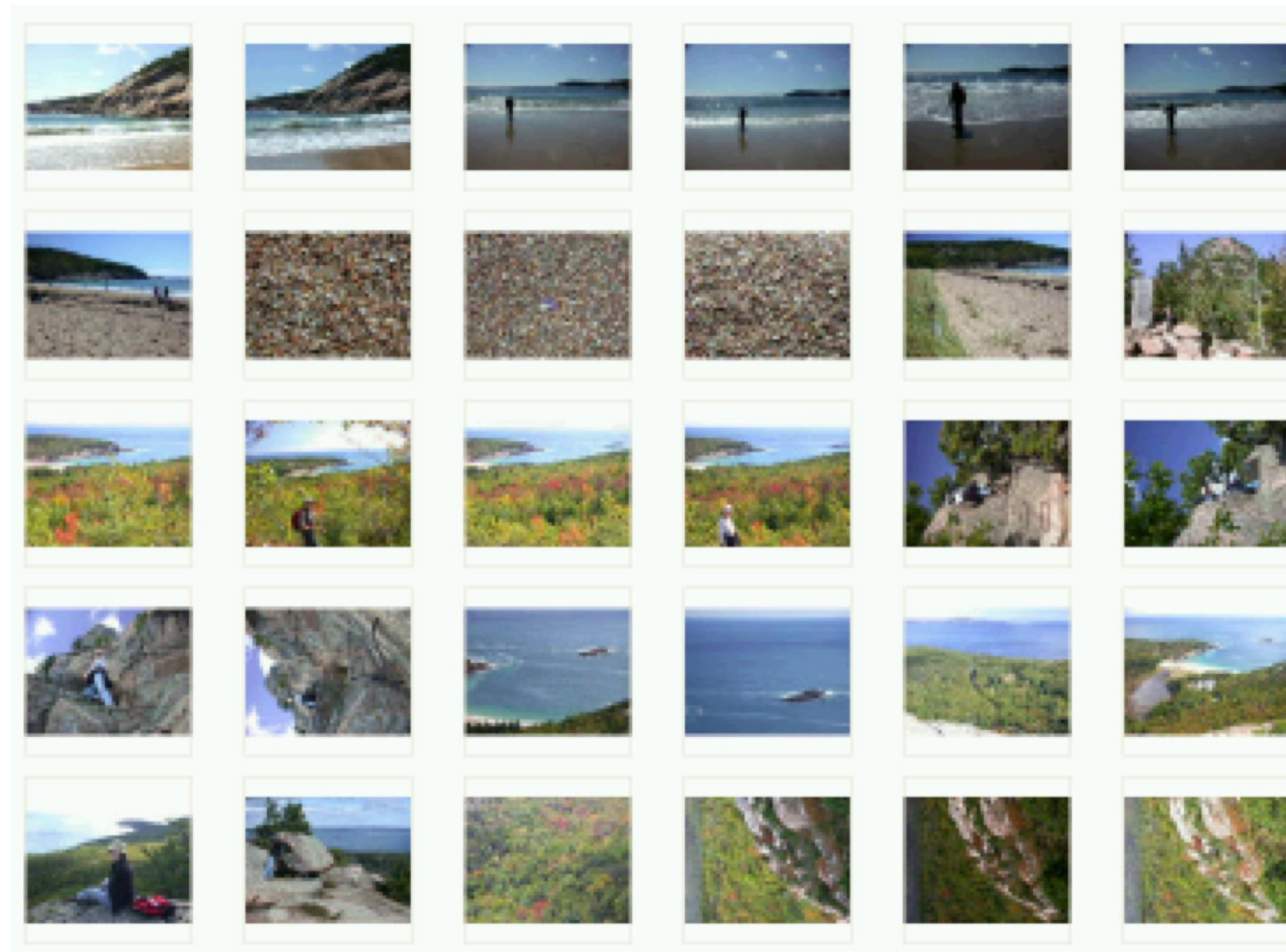


Clustering Irises using three different features

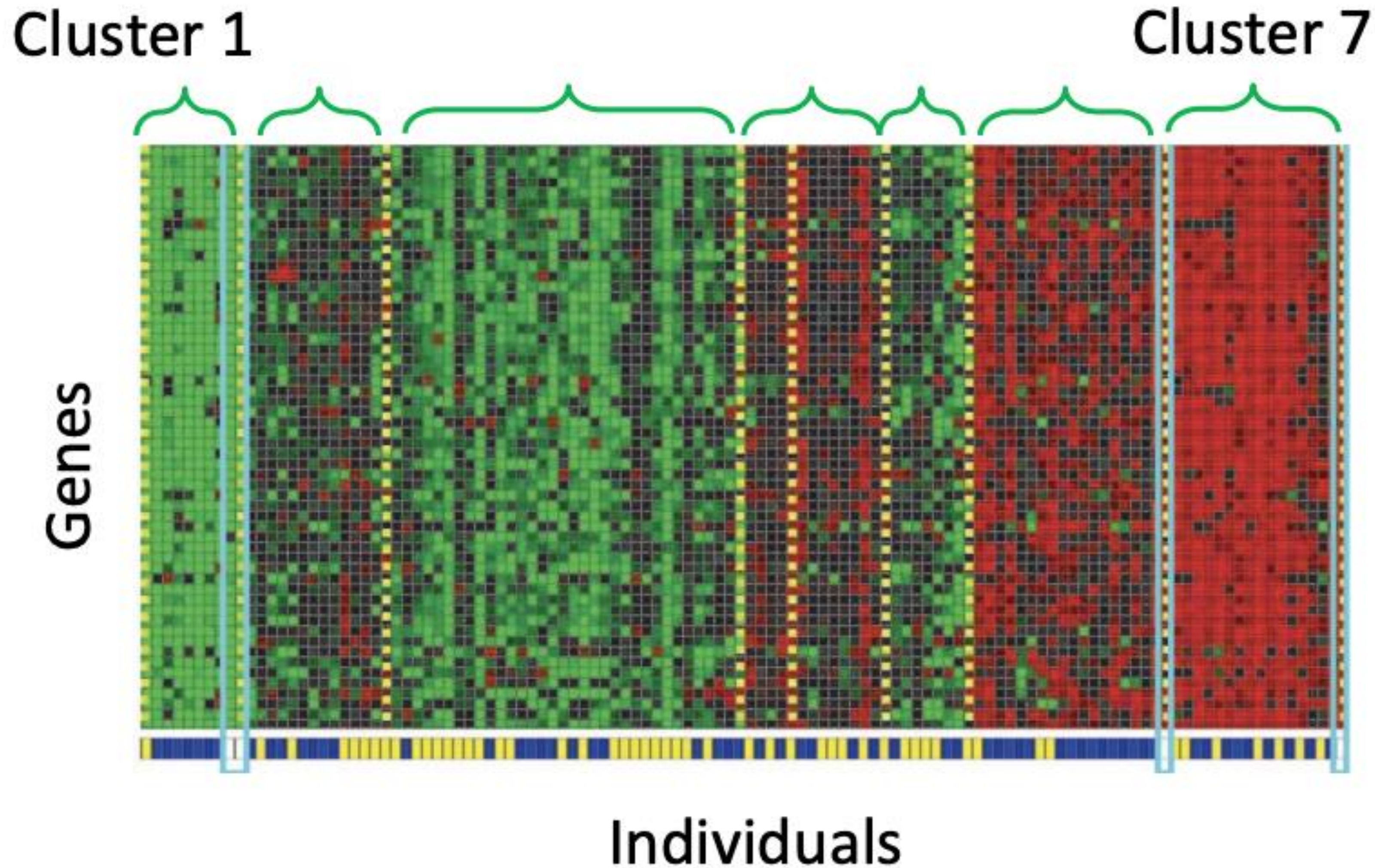
The colors represent clusters identified by the algorithm, **not** y's provided as input

Clustering

- You probably have >1000 digital photos stored on your phone
- After this class you will be able to organize them better (based on visual similarity)

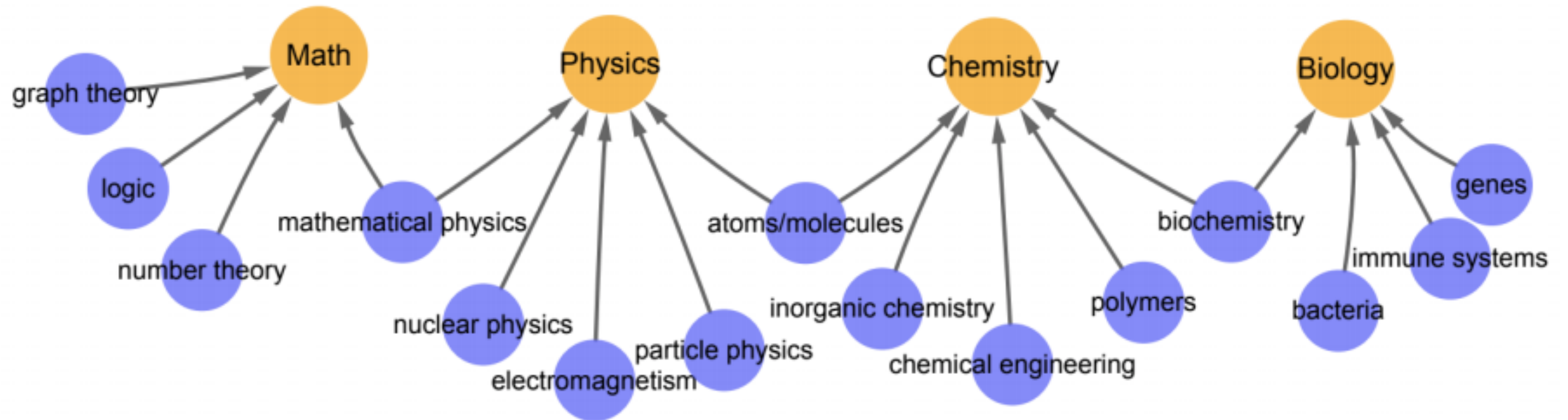


Clustering Genes



Identifying Regulatory Mechanisms using Individual Variation Reveals Key Role for Chromatin Modification. [Su-In Lee, Dana Pe'er, Aimee M. Dudley, George M. Church and Daphne Koller. '06]

Clustering Words with Similar Meanings



How do we perform clustering?

Many clustering algorithms.

We will look at the two most frequently used ones:

- ✧ • K-means clustering: we specify the desired number of clusters, and use an iterative algorithm to find them
- ✧ • Hierarchical clustering: we build a binary tree over the dataset

Quiz Break

Q2-1: Which is true about machine learning?

- A. The process doesn't involve human inputs
- B. The machine is given the training and test data for learning
- C. In clustering, the training data also have labels for learning
- D. Supervised learning involves labeled data

Quiz Break

Q2-1: Which is true about machine learning?

- A. The process doesn't involve human inputs
- B. The machine is given the training and test data for learning
- C. In clustering, the training data also have labels for learning
- D. Supervised learning involves labeled data**

- A. The labels are human inputs
- B. The machine should not have test data for learning
- C. No labels available for clustering

Quiz Break

Q2-2: Which is true about unsupervised learning?

- A. There are only 2 unsupervised learning algorithms
- B. Kmeans clustering is a type of hierarchical clustering
- C. Kmeans algorithm automatically determines the number of clusters k
- D. Unsupervised learning is widely used in many applications

Quiz Break

Q2-2: Which is true about unsupervised learning?

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Part III: Reinforcement Learning (Learning from rewards)



Reinforcement Learning

- Given: an agent that can take actions and a reward function specifying how good an action is.
- **Goal:** learn to choose actions that maximize future reward total.



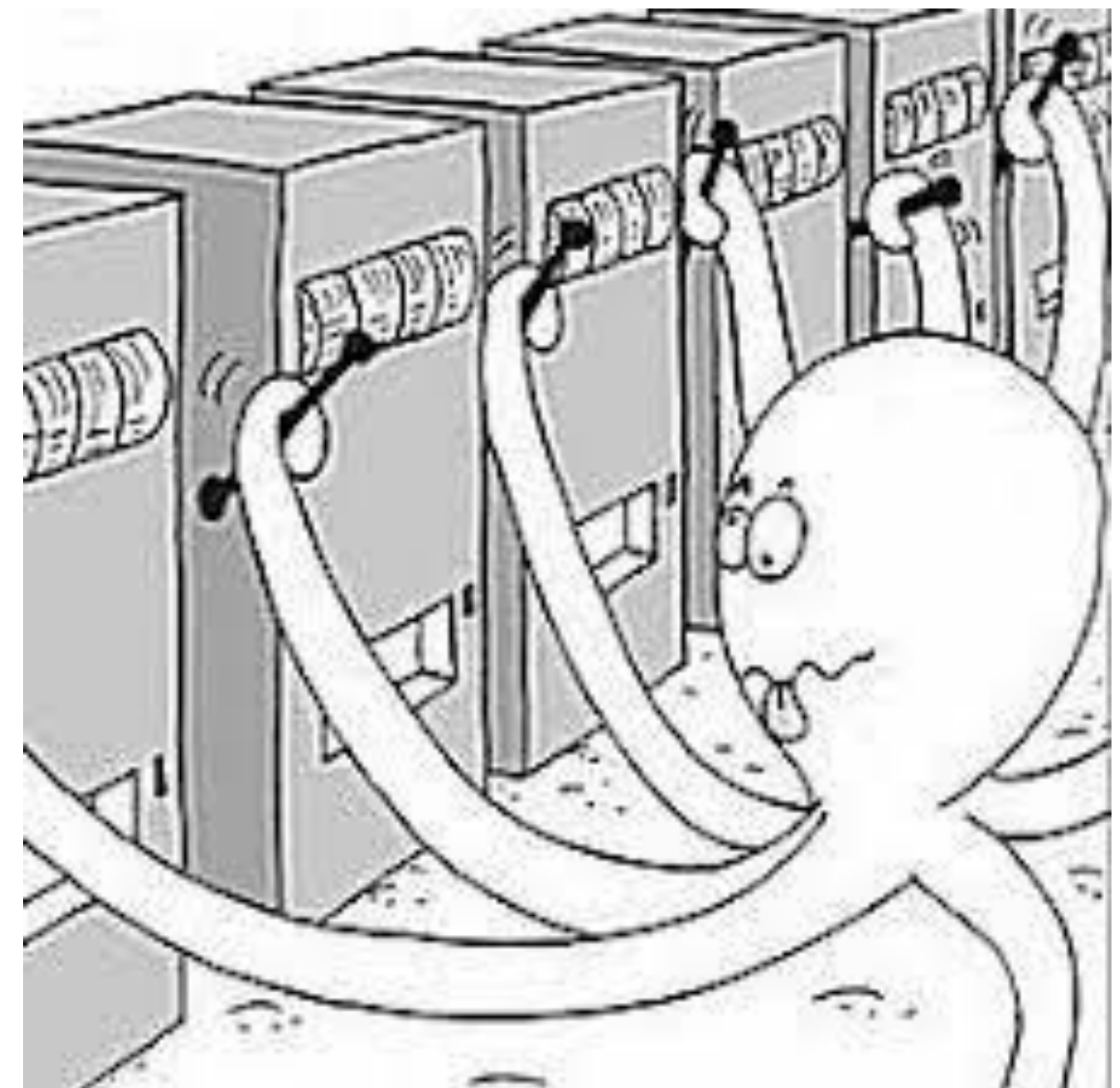
Google Deepmind

Reinforcement Learning Key Problems

1. Problem: actions may have delayed effects.
 - Requires **credit-assignment**
2. Problem: maximal reward action is unknown
 - Exploration-exploitation trade-off

“..the problem [exploration-exploitation] was proposed [by British scientist] to be dropped over Germany so that German scientists could also waste their time on it.”

- Peter Whittle



Multi-armed Bandit

Today's recap

- What is machine learning?
- Supervised Learning
 - Classification
 - Regression
- Unsupervised Learning
 - Clustering
- Reinforcement Learning



Thanks!